PARTY SYSTEMS, INFORMATION FLOWS, AND VOTE BUYING IN LATIN AMERICA

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by

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Abstract

by

Sergio Béjar

In this thesis, I will tell all that I know about Gnus. Gnus are wonderful little creatures that inhabit the center of the earth and give us wonderful and plentiful trees, dirt, and other earthly-things.

In short, we should love and cherish the Gnus. They can be very friendly, and are often mistaken for squirrels on the University of Notre Dame campus. Feed them whenever possible. If they get caught in trash cans, tip them over so that they can get out.

This abstract is going to continue on, including a few formulae, just for the sake of spilling over on to two pages so that we can see the author's name in the top right corner:

a2 + b2 = c2

E = mc2

e/m = c2

a2 + b2 = e/m

These equations, by themselves mean nothing. But to the common Gnu, they define a whole way of living. While intricate mathematical implications certainly do not inflitrate the majority of humans' lives, every Gnu, from birth, is imbued with a sense of mathematical certainty and guidance. All Gnus, great and small, feel at one with mathematics. The cute furry bit is just a scam for their calculating minds.

This dissertation will not address this subject at all, but I felt compelled to mention it here. Mainly so that we can spill over to two pages and see my name at the top of the second. Don't you think that Gnus are good things?

A Papá, Mamá y Plokis.

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PREFACE

I would like to preface this work with all the wonderful things that Gnus have brought to our society: trees, dirt, flowers, grass, lakes, and other earthly-things. We should not forget them in our daily lives.

Additionally, we should offer them food for all their hard work. In fact, Gnus work so hard that they sleep for the colder half of the year. As such, they tend to grow a little rotund. Humans should not fault them for this, as it is necessary for their survival. Indeed, many humans grow rotund on their on accord!

ACKNOWLEDGMENTS

I would like to thank Scott Mainwaring and Bumba Mukherjee, with whom this work would not have been possible. They not only encouraged me to : . Particularly the one that comes to the window in the Hayes Healy building. He (she?) has given me much inspiration, love, and dirt. I would also like to thank my advisor, Dr. Gary Greenfield, with whom this work would not have been possible.

Finally, I would like to thank the U.S. Government, Department of Gnus, for their generous grant, number GNU3042920920.3, which allowed me to pursue my work.

CHAPTER 1:

INTRODUCTION

1.1 Puzzle and Research Questions

During the past quarter-century, the most exciting trend in Latin America has been democratization. Today, all Latin American countries, except Cuba, have competitively elected governments (Mainwaring and Hagopian 2007). Several Asian countries such as Bangladesh, Nepal, the Philippines, South Korea, and Taiwan and more than 55 African countries also experienced a democratic transition during the last decade of the 20th century, and enjoy now broader and more comprehensive democratic practices. Ironically, the onset of democracy across the developing world has also created incentives for political parties and politicians to manipulate elections by distributing monetary or material benefits to individuals during elections. In other words, it has created more incentives for vote buying. [1]

Since the existence and pervasiveness of vote buying is likely to have negative consequences for democracy and economic development,[2] it is not surprising that understanding the determinants of this phenomenon has become an important area of

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1 As Schaffer (207:2) suggests, political parties in many developing countries frequently hand out not only cash, but also a variety of goods and services including chickens, rice, constructions materials, clothing, and whiskey.

2 Existing literature lacks a systematic proof of this claim. However, the academic consensus seems to support the idea that the exchange of goods for votes distorts the relationship between citizens and elected governments (Stokes 2005), thus jeopardizing the prospects of democracy. In addition, Magaloni (2006) suggests that by shortening the time-horizons of public spending policies create poverty traps that negatively affect economic development.

research among social scientists (Auyero 1999; Kitschelt 2000; Brusco et al 2005; Stokes 2005; Nichter 2008; Vicente 2008; Finan and Schechter 2009). While some scholars suggest that vote buyers (i.e. politicians) primarily target poor voters (Kitschelt 2000; Cornelius 2004; Stokes 2005; Magaloni 2006), others argue that it is the existing system of reciprocal exchange between politicians and citizens that better explains this instance of political corruption (Brusco et al 2004; Finan and Schechter 2009).[3] Both models provide important insights about the determinants of this act of political corruption. Yet when we examine the available data closely, we not only find that this phenomenon is far from being constant and homogenous across and within countries, but also, that structural, cultural, and socioeconomic factors insufficiently explain why vote buying dominates the exchange between voters and candidates over other forms of exchange (e.g. votes in exchange for pork, patronage, or policy).[4]

For example, recent surveys in Argentina and Mexico place the frequency of vote buying between 16 and 27 percent (Pfeiffer 2006; Brusco et al 2004; Stokes 2005). In addition, Abramo (2007) reports that, on average, 7% of registered voters received offers to sell their votes in the 2006 Brazilian legislative elections.[5] In other Latin American countries like Costa Rica, Peru, and Chile the frequency of vote buying ranges from 3 to 7 per cent

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[3] Vote buying has been studied using different methodological approaches. However, as Stokes (2007) suggests, most of the existing studies explain the citizen-politician relationship that emerges from vote buying using qualitative case studies that employ ethnographic techniques.

[4] I do not underscore the importance of structural, cultural, and socioeconomic factors as determinants of vote buying. But vote buying is embedded in a larger political order so political and institutional factors can explain why candidates prefer vote buying over other electoral strategies.

[5] This percentage is equal to 1,313,000 voters.

(Speck and Abramo 2001; Pfeiffer 2006).[6] Survey and anecdotal evidence suggests that this phenomenon is also prevalent in East Asia, African and the Middle-East. Schaeffer (2005), for instance, documents that, in the Philippines, more than 3 million people were offered some form of payment in the 2002 local elections. In Thailand and Taiwan,[7] mass surveys reveal that approximately 30 per cent of the voters have been offered money for their votes. In Africa, survey data from 18 nations reveal that as many as 50% of citizens in Nigeria and Sao Tome Principe have been offered bribes in exchange for their votes (Transparency International 2006; Vicente 2008, Bratton 2008).[8]

Preliminary evidence from surveys also suggests that the relationship between income and vote buying is not straightforward. For instance, in surveys conducted in the 2001 Brazilian legislative elections, Speck and Abramo (2001) find that only 8% of the persons who received monetary offers from parties or candidates had a family income of less than 2 minimum wages.[9] In the 2006 legislative elections in Brazil, 6% of the voters who were asked to surrender their vote in exchange for money had an income of one minimum wage whereas 7% of those receiving monetary offers had family incomes of more than 10 minimum wages.[10] In addition, results from a survey conducted by this author during the 2008 local elections in Nayarit suggest that the link between reciprocity and vote buying is

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6 In addition, González-Ocantos et al (2010) report that 24% of registered voters were offered a gift or service in exchange for their vote in the 2008 Nicaraguan municipal elections.

7 In Thailand, the survey was conducted by the ABAC Poll Research Institute in the 2005 general election.

8 Survey data extracted from the Afrobarometer Round 3 surveys.

9 The findings in Schaffer (2004), that presents survey evidence from the Philippines, also suggest that the link between income and vote buying is not as strong as some scholars have argued.

10 Speck and Abramo (2006) also report that 7% had family income from 1 to 2 minimum wages; 10% had a family income of 2 to 5 minimum wages; and 9% had an income of 5 to 10 minimum wages.

shaky on empirical grounds. More specifically, voters were asked: Do you think that voters who receive money from political parties feel the obligation to vote for the party and/or candidate making the offer? In all, 83% responded that voters do not feel an obligation to reciprocate the monetary offer, 10% reported that voters feel the obligation to reciprocate, and the remaining 7% did not know.[11]

Together, the examples provided in the previous paragraphs not only suggest that vote buying varies enormously across space and time, but also indicate that while cultural and socioeconomic factors can explain the existence of this phenomenon, it is embedded in a larger political order and thus political and institutional factors can also explain why candidates prefer vote buying over other electoral strategies. The examples provided above lead to the following substantive questions: Why is vote buying more frequent in some political units than in others? More specifically, can political institutions explain the frequency of this phenomenon?[12] In addition to these questions, the unfortunate fact that vote buying is a frequent and pervasive phenomenon across and within new and developing democracies suggests that we need to systematically understand the aggregate political consequences of this individual-level phenomenon. This dissertation systematically evaluates, on theoretical and empirical grounds, the political consequences of this act of political corruption during elections. I put forth an answer to the following question: to what extent

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11 The random survey designed and conducted by the author during the 2008 local elections in Nayarit, Mexico. A detailed description of this survey is presented in chapter 4 of this dissertation.

12 Ethnographic and survey evidence from developing countries suggests that vote buying not only varies in its degree but also in the forms it takes. Although analyzing the determinants of different vote buying strategies is theoretically and empirically important, in this dissertation I do not attempt to explain why politicians use one vote buying strategy over others.

do differences in the level of party system institutionalization affect aggregate electoral fraud?

1.1.1 Dissertation's Objectives

The scholarly community generally sees vote buying as an economic exchange in which voters sell their votes to the candidate who offers the higher price (Schaeffer 2005).[13] Yet, most of the existing literature overlooks the fact that vote buying is not uniform across new democracies. Therefore, they cannot explain when and under what conditions candidates are more likely to use vote buying as a campaign strategy. In addition, much of this literature focuses on theoretically explaining the supply side of this act, explaining when politicians are more likely to buy the votes of their constituencies ignoring the micro-foundations of voter and politician behavior. They overlook the potential impact of domestic political institutions on the strategic act of vote buying. My *first* research objective is to provide a comprehensive theoretical framework that explains variation in vote buying across different political units and then test the empirical validity of the answers I propose. Rather than analyzing the logistics of vote buying –that is, how candidates and parties organize vote brokers, avoid detection, and minimize slippage- or the cultural and socioeconomic, the answers provided in this dissertation not only pinpoint *how* and *why* the level of party system institutionalization triggers the candidate's desire to offer money for votes, but also how they influence the voter's decision to engage in this act.

The *second* objective of this dissertation is to propose a general framework for analyzing the potential political effects of this act of political corruption at the aggregate level

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13 In other words, the act of vote buying can be seen as a contract, or perhaps an auction in which voters sell their votes to the candidate who offers the highest price.

and to test it using an original time-series-cross-sectional dataset. Therefore, I also analyze the political consequences of vote buying. Scholars have theorized that vote buying has negative consequences in both economic (Magaloni 2006; Robinson and Verdier 2007; Baland and Robinson 2007; Stokes 2007a) and political (Piatonni 2001; Stokes 2007b) fronts. Yet they have neither provided an institutional explanation nor a systematic quantitative test of these arguments.

The *third* goal of this manuscript is to develop a multi-methodological approach that not only allows me to bridge the qualitative and quantitative research traditions in comparative politics, but also provides me with the opportunity to obtain individual-level data that will be used to systematically test a variety of hypothesis about the variation of this corrupt act. Scholars have primarily used ethnographic and survey evidence to analyze the determinants of vote buying. This dissertation, in contrast, is the first systematic attempt to explain variation in vote buying and its political consequences using a multi-methodological research design, including a field experiment, survey research, large-N time-series-cross-sectional statistical analysis, and in-depth case studies.

The rest of this chapter proceeds as follows. The next section briefly reviews existing theoretical explanations of vote buying. The third section describes the theoretical arguments that address the research questions posited above and lists the testable hypotheses that are empirically tested in this dissertation. I then explain the multi-methodological research design that I employ to empirically test the hypotheses that emerge from my theoretical argument. Section 5 provides a brief summary of how this dissertation is organized. In the conclusion of this chapter, I summarize the dissertation's main theoretical contributions and explain why a comprehensive theory of vote buying in developing countries is important.

1.2 Literature Review

Vote buying involves the distribution of money rather than ideological or programmatic appeals with the expectation of political support (Kitschelt and Wilkinson 2007). Social scientists (i.e. economists, political scientists, and sociologists) have advanced two different explanations for the existence of vote buying: the high discount and the norms of reciprocity models.

The high discount rate model suggests that poverty and constituency size fuel the demand for and susceptibility to vote buying (Kitschelt 2000, 2007; Brusco et al 2004; Stokes 2005). In this approach, the low levels of income provide politicians and voters with selective incentives that make vote buying a profitable campaign strategy. But as income increases so does the marginal cost of vote buying and thus appeal to voters using other strategies. Likewise, existing literature predicts a negative relationship between urbanization and vote buying. This is because in small communities (i.e. rural) parties can monitor votes more easily, and social relations allow parties to track how people vote therefore reducing the commitment problems associated with this act and making it a more profitable electoral strategy[14].

Other scholars suggest that it is people's feeling of obligation to return the favor that more accurately explains this practice (Lemarchand 1972; Scott 1972; Brusco et al 2004; Finan and Schechter 2009). In this line of reasoning, social norms diminish the commitment problems that arise in situations where the ballot is secret and voting is voluntary (Stokes 2005). For instance, using rural societies as example, Scott (1972) argues that norms of

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14 Vote buying becomes particularly puzzling in polities where the ballot secret and voting is voluntary. This is because political parties and candidates cannot completely enforce this act.

reciprocity reinforce clientelistic relationships in these regions. Recent work by Finan and Schechter (2009) explores systematically the hypothesis that links vote buying with reciprocity.[15] They find that vote buying is sustained by an internalized norm of reciprocity which implies that voters who receive money from candidates feel obliged to vote for the candidate/party that offered them money before the election.

A different body of literature explains why the vote buying relationship exists. Brusco et al (2004) and Díaz-Cayeros et al (2004) suggest that the voters' fear of losing future benefits determines the existence of vote buying. Others argue that it is the ability of political machines to monitor voters (Stokes 2005)[16] or the ability to turnout core supporters (Nichter 2008) that drives parties and candidates to engage in vote buying. These studies emphasize the repeated nature of clientelistic interactions since patrons (parties) and clients (voters) are often embedded in social networks that support patterns of cooperation over time.[17]

The literature mentioned above provides valuable lessons about the determinants of vote buying but suffers from three important drawbacks. First, existing studies have almost exclusively focused on explaining why the patron-client relationship embedded in vote buying exists assuming that this phenomenon will always have an effect on voter behavior. Yet the empirical support for the hypothesis that vote buying has a predictable effect on

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15 Finan and Schechter (2008) study this link using experimental and survey evidence from Paraguay.

16 Stokes (2005) develops a dynamic game theoretical model to explain how political parties mobilize voters mobilize voters through clientelistic appeals. According to Stokes, clientelistic parties minimize the informational asymmetries implicit in clientelistic exchanges by inserting themselves into the voters' social networks, and, once deeply inserted, by threatening to punish voters for voting for another party (i.e. perverse accountability).

17 Similarly, Cox & Kousser (1981) and Cox (1987) document that legal electoral reforms (i.e. secret vote) in the United States and the United Kingdom played a decisive role in reducing electoral corruption. They argue that vote buying had perverse effects on voter turnout.

voters is limited and mixed. For example, Lindberg and Morrison (2008) find that voters in Ghana evaluate candidates based on policy platforms rather than on ethnic and clientelistic bases. Similarly, Young (2009) suggests that vote buying has no effect on the incumbent's vote share. By contrast, Bratton (2008) and Vicente (2008) argue find that vote buying has important effects on voter behavior in Nigeria and Sao Tome Principe, respectively.[18] The lack of empirical support for the hypothesis linking vote buying with voter behavior suggests that we need to develop a more systematic understanding of the micro-foundations of voter's and politician's behavior.

Second, vote buying is a widespread and persistent phenomenon in many new and developing democracies. Yet, as mentioned before, it varies tremendously across space and time. The substantial variation in vote buying across different political units suggests that this phenomenon is a strategic rather than a random act not only on the part of political parties and candidates but also on the part of voters. In other words, vote buying does not happen in a vacuum. Even more puzzling is the fact that scholars have overlooked that the characteristics of some political institutions (party systems, for instance) may shape one's decision to exchange votes for money.[19] Political institutions, as Gourevitch (2008) suggests, are likely to shape the strategic decisions of political actors at the individual-level and, consequently, affect their behavior.

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18 The effects of vote buying on voter behavior in regions other than Africa are very limited.

19 Hicken (2007) is among the first scholars to suggest that institutions like electoral systems or competitiveness of elections explain the existence of vote buying. His focus, however, is on intra-party and not inter-party competition.

The existing literature has paid very little attention to the aggregate political consequences of vote buying.[20] Scholars have theorized that democracy and economic development are likely to suffer when this phenomenon is present. Although intuitively appealing, these theories tell us little about the mechanism through which an individual-level phenomenon such as vote buying affects aggregate political outcomes. In other words, they fail to provide a comprehensive theory explaining how and why the aggregation of individual-level preferences affects a macro-level phenomenon like democracy. Moreover, existing studies have not yet systematically tested their predictions.[21]

While existing explanations of vote buying have enhanced our understanding about this act of political corruption, they paint with a very broad brush its micro-level foundations and thus the incentives that shape this phenomenon. This dissertation fills an important theoretical gap in our understanding of vote buying within and across developing countries in two fundamental ways. First, by developing a comprehensive theory that explains how and why an institutional variable (i.e. the level of party system institutionalization) affects the strategic behavior of politicians (supply-side) and voters (demand-side), this study will help us understand why vote buying is more likely to occur in some regions than in others, thus providing us with a better understanding of the citizen-politician relationship in developing countries. Second, this manuscript pushes forward the research on the political consequences of vote buying, shedding light on the link between vote buying and an aggregate political outcome like electoral fraud.

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20 Yet there is not systematic empirical evidence supporting the results of these models.

21 As I explain below, this work provides a systematic empirical test for the consequences of vote buying.

Methodologically, my dissertation contributes to the existing literature on the determinants and consequences of vote buying: this research project is the first attempt to explain variation in vote buying and its political consequences using a multi-methodological research design, including in-depth case studies, a field experiment, survey research, and large-N time-series-cross-sectional statistical analysis. A multi-methodological approach bridges the qualitative and quantitative research traditions in comparative politics, and it also allows me to carefully analyze the validity of the hypotheses that emerge from my theoretical argument.

1.3 Theoretical Argument and Testable Hypothesis

Social science still needs a theory that explains the variation in vote buying practices across and within developing countries, and the political implications of this instance of political corruption. In this section, I discuss the main theoretical framework, causal arguments, and testable hypotheses that I propose to answer the research questions that motivate this dissertation.

1.3.1 The Micro-level

What explains variation in vote buying? To answer this question, I rely on existing models of clientelism and vote buying (Dixit and Londregan 1996; Stokes 2005; Díaz-Cayeros et al 2007), but substantially extend them by suggesting that differences in the level of electoral uncertainty that result from the degree of party system institutionalization can

explain why vote buying is more frequent in some regions than others.[22] The theory developed in this manuscript posits that the level of party system institutionalization affects both the politicians' and voters' strategic decision to engage in vote buying. More broadly, my theory links the level of party system institutionalization to the frequency of vote buying, and to the citizens' rational decisions to exchange money for votes.

I argue that the unstable patterns of political competition engendered by weakly institutionalized party systems explain why vote buying occurs more frequently in some political units than in others. High levels of electoral volatility are likely to (i) generate important changes in the composition of the electorate (Madrid 2005, Zielinski et al 2007), and (ii) increase the competitiveness of elections (Lachat 2009). These effects, in turn, not only create a market for votes,[23] but also introduce a high degree of uncertainty into the candidates' decision process.[24]

When the level of electoral uncertainty is high, politicians and political parties must modify campaign strategies and invest their resources in the segment of voters who make the difference between electoral victory or defeat. Rather than investing on core supporters, politicians will focus their attention on obtaining the electoral support of swing voters.[25] Floating voters are unlikely to be attentive to the political cues emitted by political parties,

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22 As mentioned before, Hicken (2007) analyzes the effects of intra-party (as opposed to inter-party competition) to explain vote buying. By contrast, I am interested in analyzing the effects of inter-party competition.

23 High levels of electoral volatility and weak parties reduce the voters' attachments to political parties. This, arguably, reduces the number of core supporters and increases the proportion of swing or floating voters thus creating a market for votes.

24 This is because in highly competitive elections it is very difficult to correctly predict the outcome of the election.

25 This is because swing voters create the market for votes.

suggesting that as the number of marginal votes that ensure electoral victory increase so does the costs of appealing to those voters using programmatic appeals. Hence, rather than spending on campaign advertising, politicians aim at developing private and personal relationship with the electorate, and offer money for their votes thus increasing the likelihood of vote buying.

By contrast, when the levels of electoral volatility are low and parties have stable roots in society, vote buying is not a cost-effective electoral strategy for parties and candidates. Under these circumstances, voters are likely to cast their votes on programmatic or ideological bases.[26] In addition, low levels of electoral volatility substantially decrease the level of electoral uncertainty. In these conditions, the price of each vote can be substantially higher given that there are only a very limited number of votes for purchase, thus decreasing the marginal utility of vote buying. As the marginal utility of engaging in this corrupt act decreases, candidates see vote buying as a less profitable campaign strategy thus reducing its likelihood. The preceding discussion leads me to state the following hypothesis:

Hypothesis 1: The higher (lower) the level of party system institutionalization, the more (less) vote buying will occur.

I also suggest that the degree of party system institutionalization affects the voters' decision to sell their votes. Specifically, the high degree of electoral uncertainty that is common in weakly institutionalized party also affects the citizens' electoral behavior (Sartori 1986; Moser 2001). Fluid party systems not only provide voters with limited information

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26 Specifically, ideology serves as a shortcut that allows voters to minimize the costs of obtaining political information. The effects of ideology and party identification have been extensively documented by political scientists in developed democracies. For example, Whitten and Palmer (1996) suggest that in Britain and the Netherlands, which arguably possess high levels of party system institutionalization, ideology and party identification substantively influence voting behavior.

cues about the election, but also with poorly developed sources of political information (Moser 2001). Therefore, voters in countries with these systems are often uncertain about the utility of voting and unaware that vote buying can have damaging consequences for democracy.[27]

The low levels of political information and awareness that are common of citizens in weakly institutionalized party systems affect their decision to exchange votes for money as follows. First, given that citizens form political attitudes using available information, the lack of information about vote buying not only increases their uncertainty about the frequency of this phenomenon, but also with respect to its potential consequences. Since information flows allow citizens to link attitudes with specific issues, the lack of developed sources of political information, in turn, has two effects on the voters' predisposition toward vote buying. On one hand, it decreases its salience in the public debate, creating an environment that diminishes the citizens' general understanding about its frequency and consequences. As the understanding of the frequency and consequences of vote buying decreases, citizens are more likely to be indifferent or even accept this practice. On another, it generates a great deal of uncertainty about the future benefits of voting based on ideological or programmatic appeals so they rapidly discount promises of future benefits making vote selling attractive to them (Hicken 2002). Following this line of reasoning, I formulate the following hypothesis:

Hypothesis 2: The lower the level of political information, the more likely voters are to perceive that the frequency of vote buying is low.

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[27] Survey evidence from Mexico and Nigeria confirm that a high proportion of voters are unaware of the consequences of vote buying. For example, in Mexico, voters in Nayarit

The lack of political information makes citizens more likely to sell their votes. Two reasons explain this point. First, it makes it more difficult for citizens to develop a set of rules against this act of political corruption. In the absence of information about the potential negative consequences of vote buying, voters are unlikely to understand the consequences of selling their votes. Therefore, the likelihood of developing a revulsion against this act is very low. Second, it prevents citizens from learning and internalize norms against this phenomenon. This, in turn, makes it difficult for voters to process and understand why rejecting a monetary offer from a political party and/or candidate makes sense. Since voters not only do not have a clear understanding about the frequency and harmful consequences of this phenomenon, but also cannot develop a set of rules against it, they are more likely to engage in it. Thus, the third hypothesis is:

Hypothesis 3: The lower the level of political information, the more likely voters are to sell their votes.

1.3.2 The Macro-level Hypothesis

The three hypotheses stated in the preceding discussion uncover an interesting relationship between the level of party system institutionalization and vote buying, but do not shed light on another relevant question: does the micro-level vote buying behavior have any consequences for aggregate political outcomes?

That election fraud contaminates elections in both developed and developing countries is well-know. Alvarez et al (2008) document, in 2006 alone, that presidential elections in Mexico, Italy and several former Soviet republics were surrounded by significant allegations of electoral fraud. The frequency and salience of this modality of political corruption has motivated political scientists to formulate explanations for its existence.

While some existing studies have found that electoral fraud is correlated some socio-economic factors (Eisendstadt 1999; Trejo and Aguilar-Rivera 2002; Lehoucq and Molina 2002; Alston and Gallo 2009; Ziblat 2009), others have suggested that the observed variation in electoral fraud can best be explained by political competition (Simpser 2005),[28] or other institutional factors such as electoral rules (Lehoucq and Molina 2002). While these factors may encourage politicians to rig elections in developing countries, they pay little attention to the micro-foundations of electoral corruption.[29] By contrast to existing explanations of electoral fraud, I argue in this dissertation that by increasing the likelihood of vote buying at the individual level, weakly institutionalized party systems trigger the incumbents' desire to use electoral corruption to hold their offices.[30]

In the previous section I suggested that the unstable patterns of electoral competition together with the important lacunas of political information that are common in fluid party systems affect the politicians and voters strategic behavior in such a way that they create a market of "swing" voters in which vote buying could be a profitable electoral strategy. Although the existing theoretical literature on vote buying emphasizes that vote buying is generally practiced by incumbents, the empirical and anecdotal evidence suggests that both incumbent governments and opposition increasingly rely on vote buying to influence the outcome of elections.[31]

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28 With some important exceptions, Simpser (2005) finds a positive correlation between competitiveness and various measures of electoral fraud.

29 I use the terms electoral corruption and electoral fraud indistinctively.

30 I assume that survival and longer time-horizons in office maximize the incumbents' utility functions.

31 For instance, a survey conducted by the author during the 2008 local elections in Nayarit, Mexico reveals that more than 50% of the monetary offers received by voters in the 2008 local elections came from

The fact that parties that vote buying is practiced not only by incumbent parties and candidates but also from the opposition has, in turn, one fundamental consequence on the likelihood of electoral fraud. Vote buying is an individual-level phenomenon in which both candidates and voters participate. Given its illegal nature, proving its existence is challenging as neither politicians nor citizens have incentives to truthfully cooperate against organizers of such schemes. Therefore, allegations of vote buying are difficult to prosecute and punish (Alvarez et al 2008) and candidates from all parties face increasing incentives to buy votes.

By resorting to vote buying, opposition parties and candidates not only increase the level of competitiveness of the election but also have credible chances to flip the outcome of the election in their favor. This is because, as mentioned before, the market of votes in countries with weakly institutionalized party systems is likely to have a majority of floating voters who can effectively be swayed away with money or other material goods. Therefore, widespread vote buying increases the uncertainty about the incumbent's prospects of survival in office. Since retaining office maximizes the incumbents' utility function, an increase in electoral uncertainty triggers a corrupt response from the incumbent government which, in turn, increases the likelihood of electoral fraud.[32]

More specifically, an increase in electoral uncertainty can trigger at least two types of corrupt electoral behavior from incumbent governments. On one hand, incumbents can indiscriminately use vote buying as a means to secure an electoral victory and play a "blame-game" in which the opposition can also be easily accused of corrupting elections. The

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opposition parties. A report from The Philippine Daily Inquirer also suggests that both incumbents and opposition bribed voters in the 2010 local elections in the Philippines

32 Please recall that appealing to voters on the basis of ideological or programmatic appeals is not profitable in countries with weakly institutionalized party systems.

pervasive use of vote buying at the individual-level can thus sum-up in the aggregate, increasing the probability of electoral fraud. On another, incumbent governments can also respond to an increasing electoral uncertainty by manipulating the election secretly placing fictitious names on the voter rolls, casting bogus votes in the names of persons who did not vote, obtaining and marking absentee ballots without the active input of voters involved, and falsifying vote tallies. Based on this discussion, I state the following hypothesis:

Hypothesis 2.4: The lower (higher) the level of party system institutionalization, the higher (lower) the likelihood of electoral fraud.

1.4 Research Methodology

I adopt a multi-methodological to test the hypotheses that emerge from my theoretical argument. This multi-methodology approach includes comparative case-study research, survey data analysis, a field experiment, and tests on an original large-n time-series-cross-sectional dataset (TSCS hereafter). I briefly describe this multi-methodological strategy for my empirical tests below.

The theoretical argument developed in this dissertation generates four testable hypotheses that were summarized above. The first three hypotheses predict the effects of party system institutionalization and information flows on vote buying. The main predictions of these hypothesis are: (i) that vote buying is more frequent in countries that have inchoate party systems, and (ii) that the lack of information about the negative consequences of vote buying increases the probability that voters will accept money in exchange for goods. Testing the first hypothesis, however, poses an important challenge as obtaining reliable information about this instance of political corruption is not an easy task.

To circumvent this problem, I combine extensive qualitative evidence and survey methodology from Mexico to analyze the predictions posited by hypothesis 2.1.

The hypotheses that link information flows with vote buying (H 2.2 and H 2.3) predict individual-level behavior. Therefore, an appropriate empirical research design requires the use of a randomized field experiment. Although field experiments are very useful to test theories at the individual level, they are not a common research strategy in comparative politics.[33] From a methodological perspective, the principal virtue of field experiments is that randomization provides a well-defined procedure for deriving unbiased assessments of causal inference. The estimations of field experiments are likely to be unbiased for two fundamental reasons. First, in randomized field experiments the independent variable is the only factor that varies systematically in the experiment. In other words, if the experiment is appropriately controlled, the confounding variables are eliminated. Furthermore, field experiments allow researchers to develop a dependent variable that truly reflects the phenomenon under study (i.e. internal validity), and to minimize the possibility of measurement error (i.e. that the variable can be measured accurately). I thus carefully designed and conducted a randomized field experiment during the 2008 local elections in Nayarit, Mexico. Results from the three-month field experiment in 9 local electoral districts of the aforementioned Mexican state provide strong support for my theoretical predictions.

The last hypothesis examines the impact of party system institutionalization on electoral fraud across Latin American countries. In particular, the main claim is that by increasing the likelihood of vote buying at the individual level, low levels of party system

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33 Wantchekon 2003; Guan & Green 2006; Humphreys et al 2006.

institutionalization increase the likelihood of aggregate electoral fraud in these democracies.[34] I statistically test this claim in an original time-series-cross-sectional dataset of observed elections in 18 Latin American countries between 1982 and 2006.[35] Results from a variety of logistic models provide strong statistical support for the aforementioned hypothesis and remain robust to different specifications of the model and different econometric estimators. Therefore, the results obtained in this large-n analysis allow me to be confident about the potential generalization of my results in both its aspirations and empirical grounding[36]

1.5 Plan of the Dissertation

This dissertation is organized as follows. Chapter 2 presents a comprehensive (demand and supply sides) theory of vote buying and the testable hypotheses of this dissertation. The theory laid out in this chapter answers two fundamental research questions. Namely, what explains variation in vote buying across developing countries? And to what extend does vote buying at the individual-level affect an aggregate political outcome such as electoral fraud? These questions are motivated by puzzles about democracy in Latin America, and they are relevant to other regions of the world.

In Chapter 3, I present the quantitative tests for hypothesis 2.1, 2.2, and 2.3. The chapter first presents the results of a multiple-level logit analysis on the relationship between

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34 As I carefully explain later, the level of party system institutionalization is operationalized as the level of electoral volatility for each legislative election in each country.

35 These countries are: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, the Dominican Republic, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay, and Venezuela. The lack of democratic elections in Cuba prevented me from including this country in my sample.

36 Coppedge (2002:7) suggests that one of the main advantages of regression analysis is that it is relatively general in both its aspirations and in its empirical grounding.

electoral volatility and vote buying at the legislative level in Mexico. It also presents the results of an original randomized field experiment used to test the statistical validity of hypotheses 2.2 and 2.3.

Chapter 4 presents the statistical analysis on a large-n time-series-cross-sectional dataset, which is used to test the hypothesis linking party system institutionalization with electoral fraud in Latin America. This chapter first provides a brief explanation of the sample used in the regression analysis. I then present a detailed description of the operationalization of the dependent variable (electoral fraud), the independent (electoral volatility), and control variables. I also explain the statistical methodology employed in this analysis. I then provide a thorough description of the main results. In explaining the statistical results, I particularly emphasize about the substantive effects of my main independent variable on the dependent variable. Finally, the chapter presents a battery of robustness tests that allow one to control for different specifications of the model (i.e. alternative hypotheses), and potential econometric problems. The statistical results unambiguously show that low levels of party system institutionalization substantively increase the likelihood of electoral fraud across Latin American democracies between 1982 and 2006.

Chapter 5 presents an in-depth case study of the effects of electoral volatility on the politicians' and voters' incentives to use vote buying as an electoral strategy. In this chapter, I focus on the politics of vote buying in Mexico. Given that Mexico experienced not only a transition to democracy but also important changes in the composition of its party system and electoral competitiveness in 1997, the Mexican case allows me to carefully evaluate how the change in the party system and in the level of electoral competitiveness affect the strategic electoral behavior of political parties and candidates before and after 1997.

Chapter 6 contains the conclusions. It begins by presenting the theory and the findings of this dissertation. In addition, it explains the theoretical, empirical, and practical contributions of this dissertation not only for the literature on clientelism and vote buying, but also for extant studies on party systems of developing and new democracies. This chapter ends by outlining the implications that this study may have for: (i) the links between politicians and citizens, and (ii) the prospects of democracy in Latin American and other developing countries.

1.6 Why is Studying Vote Buying in Developing Countries is Important?

The analysis presented in this dissertation has four main contributions to the literature on vote buying. First, existing literature assumes that vote buying is constant across different political units (Brusco et al 2004; 200Stokes 2007; Nichter 2009). But, as mentioned before, the empirical evidence from developing countries suggests that vote buying is far from being a constant phenomenon. The theoretical arguments and empirical analysis in the following chapters take these studies one step forward by examining variation of vote buying across different political units.

Second, the demand-side and supply-side explanations in the literature on vote buying often talk past each other. This limits the explanatory power of such analyses and leaves us with incomplete answers about the causes and potential consequences of this act. Unlike extant studies, my dissertation brings together the supply (politicians) and demand (voters' incentives) sides of vote buying providing us with a more comprehensive picture of an act of political corruption that corrodes democracy across the developing world.

Third, existing scholarship has devoted little attention to examining the political and economic consequences of this phenomenon. Doing this is important because, as Stokes

(2007) notes, vote buying can undermine the autonomy of vote sellers endangering the prospects of representative democracy in many new and developing countries. By contrast to previous studies, the theoretical argument and empirical tests of this dissertation systematically analyze how and to what extend the incidence of vote buying at the individual-level affects the likelihood of electoral fraud and, in turn, on the prospects for democracy and its survival. Finally, as shown in the dissertation's empirical chapters, I put together several survey and time-series-cross-sectional datasets to test the hypotheses that emerge from the theoretical argument. The results from testing my claims across different types of datasets provide strong support for the hypotheses and claims of my theoretical argument.

This dissertation has also important implications for the literatures on parties and party systems and on electoral corruption. Scholars in comparative politics have theorized about the potential consequences of inchoate party systems on democracy (Mainwaring and Scully; Mainwaring and Torcal 2005).[37] Yet we still lack a systematic empirical test supporting this argument. This manuscript not only provide a comprehensive theory about the effects of party systems on vote buying (individual-level) and electoral fraud (aggregate-level), but also is the first work that systematically tests this relationship. The theoretical argument and empirical tests (chapter 5) also advance our understanding on the determinants of electoral fraud. In particular, the argument developed here indicates that by fostering vote buying at the individual-level, low levels of party system institutionalization have substantial effects on the likelihood of aggregate electoral fraud thus dampening democracy.

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37 Political scientists have also comprehensively studied the determinants of party system institutionalization in new and developing democracies (Mainwaring and Scully 1995; Mainwaring 1998, 1999; Roberts and Wibbels 1999; Mainwaring and Zoco 2007; Hicken 2008).

The findings presented in this analysis are not a mere academic exercise. Indeed, they have important two important real world implications. First, as the results presented in chapter 5 show, by increasing the likelihood of vote buying at the individual-level, inchoate party systems substantially increase the incumbent's desire to rig elections in order to maximize his/her survival in office. This, in turn, could threaten the survival of democracy in these countries.[38]

In the policy realm, governments, citizens, and other organizations interested in curtailing vote buying can also benefit from the findings of this analysis. Reports of vote buying during electoral periods in the developing world are far from being an isolated phenomenon. From Mexico to Argentina, and then to India and the Philippines the aforementioned anti-democratic practices abound. What can governments and other organizations do to prevent vote buying? If the potential negative implications of vote buying are indeed confirmed, policy-makers could aim at introducing electoral reforms that raise awareness and change voter attitudes toward undemocratic electoral practices. In addition to specific laws on vote buying, the results presented in this dissertation suggest that governments and electoral authorities could also generate hostile environments to vote buying through educational campaigns.

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38 Vote buying biases the balance of power toward the rich elite, thus undermining the capacity of the poor and the middle class to enjoy the benefits of democracy. Inability to enjoy the advantages of a full-fledged democratic regime may, in turn, produce an increasing dissatisfaction with democracy among the poor and the middle classes which could threaten the viability of democracy.

CHAPTER 2:

PARTY SYSTEMS, INFORMATION FLOWS AND VOTE BUYING: A COMPREHENSIVE THEORY

2.1 Introduction

In this chapter, I develop a theoretical argument that links party system institutionalization with the variation of vote buying in developing countries. I explained in the introductory chapter that vote buying is unfortunately a frequent and pervasive phenomenon across developing countries. Yet the forms it takes, its degree, and its political and economic consequences vary enormously across space and time. What explains variation in vote buying across new and developing democracies? And to what extent does vote buying affect aggregate political outcomes? These are the questions I explore in this chapter.

My theoretical argument suggests that differences in the level of party system institutionalization can explain both the supply and demand sides of vote buying, allowing us to better understand why vote buying is more likely to occur in some countries than others. More specifically, I argue that the unstable patterns of political competition that result from low levels of party system institutionalization increase politicians' uncertainty about the electoral outcomes. Given that voters in the developing world often have low levels of political awareness, appealing to them using programmatic cues is very costly. Hence, rather than spending on campaign advertising, politicians aim at developing private and personal relationship with the electorate, offering money in exchange for votes. Information flows

explain why some voters exchange votes for money while others do not. Specifically, I argue that information about the harmful consequences of vote buying sensitizes people to this act of electoral corruption, allowing them to learn about its frequency, severity, and potential harmful consequences. More specifically, I hypothesize that greater exposure to information about the negative consequences of vote buying makes voters less likely to accept monetary offers from political parties and/or candidates.

I also suggest that the higher incidence of vote buying resulting from low levels of party system institutionalization has important negative consequences for the prospects of democracy in developing and new democracies. Since the likelihood of vote buying by all political parties and candidates is higher in countries with fluid party systems than in those with stable systems, incumbents in the former polities may face a credible threat to their survival in office from the opposition. Thus, to maximize their political survival and increase their time horizons in office, incumbent governments in countries with inchoate party systems face higher incentives to rig the election.

My theory highlights the effects that political institutions (i.e. the level of institutionalization of the party system) can have on the existence of an act of political corruption that corrodes elections across developed and developing countries. It is a contrast to existing studies suggesting that vote buying is primarily caused by poverty (Scott 1972; Kitschelt 2000' 2007, Brusco et al 2004; Stokes 2005), constituency size (Kitschelt 2000), reciprocity (Finan and Schechter 2009, or intra-party competition (Hicken 2002; 2007) It also differs from previous studies since it not only explains why politicians and/or political parties rely on vote buying to obtain the electoral support of the citizenry (supply-side), but also what are the micro-foundations of vote selling behavior (i.e. the demand side of vote buying). Finally, it deepens our understanding about this act of political corruption by

theorizing about one of its potential political consequences: the likelihood of aggregate electoral fraud. As such, the theory presented in this chapter is comprehensive as it provides detailed hypothesis not only for the supply and demand sides of this corrupt practice, but also for its political consequences.

The rest of this chapter is organized as follows. I first set up the puzzle that motivates this chapter. I then develop a theoretical argument that explains how and why low levels of party system institutionalization affect the likelihood of vote buying in developing countries. After discussing the causal mechanisms that fuel the supply and demand-side of vote buying, I extend the argument to analyze the effects of party system institutionalization on electoral fraud. In the conclusion, I summarize my main causal arguments and theoretical contributions and then systematically list the testable hypothesis emerging from my theoretical argument.

2.2 Setting the Puzzle

E.E. Schattschneider argued more than half a century ago that political parties are central to any well-functioning democracy.[39] Among many other functions, parties allow politicians to gain office through competitive elections, help articulate group aims, nurture political leadership, develop and promote policy platforms, provide voters with coherent electoral alternatives, ensure that voters have significant electoral choices, and that those choices are translated into decisions in the policy realm. Parties do not perform these

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40 Elmer E. Schattschneider, Party Government (New York: Farrar and Rinehart, 1942).

functions in a vacuum. Rather, they aggregate and form a system which provides them with rules and patterns of competition among them (i.e. party system).[40]

But empirical evidence from new and developing democracies suggests that party systems are neither homogeneous nor remarkably stable across these countries.[41] Maiwaring and Zoco (2007), for instance, document that the mean electoral volatility in developing democracies ranges from a low 12.5 in Colombia to a substantively high 59.2 in Ukraine.[42] The substantial variation in levels of stability reported by the aforementioned scholars is not limited to Latin America and Eastern Europe. In Africa, Kuenzi and Lambright (2001) find that the mean electoral volatility goes from 3 in Malawi to 99 in Lesotho. [43,44] The finding about the unstable dynamics of party systems across developing countries is important because it can substantially influence the nature and intensity of political campaigns and raise the stakes of the electoral game.

Therefore, it is not surprising that parties and candidates in developing countries, who are assumed through this argument as actors whose only objective is to secure electoral victories,[45] have devised campaign strategies that revolve around distributing particularistic

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41 I follow Mainwaring (1999) who defines party system as the set of patterned interactions in the competition among parties.

42 For instance, Coppedge (1998) finds that electoral volatility in Latin American countries is three times higher than in advanced industrialized democracies.

43 Mainwaring and Zuco (2007) present estimations of electoral volatility in 47 countries. Their sample includes both developed and developing (Latin America, Eastern Europe and Asia) democracies.

44 These estimates are the average of electoral volatility in both legislative and presidential elections. Kuenzi and Lambright (2001) report estimates for 30 African countries from 1980 to 1999. The mean of electoral volatility across African countries is 29.44

45 Hicken (2006) also finds that party systems in Thailand, Indonesia and Philippines are weakly institutionalized.

45 In making this assumption, I follow Díaz-Cayeros et al (2007).

benefits to individuals rather than on ideological of programmatic basis. Numerous studies have extensively documented that vote buying, defined as any apparent efforts by parties or candidates to purchase political support at the ballot box through the delivery of material goods or cash (not promises) to voters, is an instrument widely used by parties in these countries to influence the outcome of elections (Hicken 2002; Brusco et al 2004; Schaffer 2006; Stokes 2007; Nichter 2008; Nitcher et al 2010).[46] In the 2002 elections in the Philippines, for example, about 7 percent of all adults allowed to vote were offered some form of payment in exchange for their vote. In Thailand, 30 percent of the heads of households revealed that they have been offered money during the 1996 election (Hicken 2002). Buying votes has also a long tradition in other developing countries such as Argentina, Mexico, Senegal, Taiwan, Nigeria, and Lebanon among others (Pfeiffer 2006; Schaffer 2007; Bratton 2008; Wang and Kurzman 2007).[47] Indeed, as the Media Line News Agency reported before the 2009 general elections in Lebanon:

> "…political parties are strenuously denying giving money or services in exchange for votes, although analysts say vote buying has been the norm in previous election campaigns."[48]

The preceding discussion raises three interesting questions. Namely, is it possible to draw a connection between the instability of party systems and the widespread use of vote

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46 In my view, vote buying has four core attributes that allow one to differentiate it from other clientelistic strategies: timing, guiding distinction, linkages, and actors. First, vote buying generally occurs within reasonable proximity to elections. Second, it is a private exchange of private (as opposed to public) goods for votes at the individual level (Stokes 2007). Third, it does not involve repeated interactions between candidates and voters over time thus resulting in unstable patterns of cooperation over time. Fourth, it is a political strategy that can be used indistinctively by either incumbent or opposition candidates in any given election.

[49] The Jerusalem Post, June 1, 2009. Available online at: http://www.jpost.com

buying across developing countries? And, if so, how? Moreover, does this micro-level corrupt political strategy have any consequence in the aggregate? In what follows, I argue that weakly institutionalized party systems modify the strategic behavior of both candidates and voters in such a way that vote buying becomes a profitable electoral strategy. Additionally, I suggest that, in the aggregate, the widespread use of vote buying incentivizes incumbent governments to rig elections thus increasing the likelihood of electoral fraud.

2.3 The Micro-Level: Party Systems and Vote Buying Strategic Behavior

2.3.1 Supply-Side of Vote Buying

The phenomenon of vote buying in developing democracies, which has adverse consequences on democratic consolidation, has motivated social scientists to explain the determinants of this phenomenon. The bulk of existing studies suggest that vote buying is primarily fueled by demand-side factors such as poverty, constituency size (Cornelius 2004; Stokes 2005; Magaloni 2006), or reciprocity (Brusco et al 2004; Finan and Schechter 2009). Both the high discount rate and the norms of reciprocity model provide important insights about the causes of vote buying. However, less attention has been paid to the possibility that institutional factors may fuel both the supply and demand sides of this act of political corruption.[49] Moreover, as mentioned before, previous studies overlook the fact that this common act of political corruption occurs more frequently in some countries than in others.

[49] To the best of my knowledge, only a handful of studies have explained the institutional determinants of vote buying (Hicken 2002; Hicken 2007). However, they primarily explain how intra-party competition motivates candidates to buy votes. By contrast, the argument presented in this chapter studies the effects of inter-party competition on vote buying.

I fill this lacuna by presenting a comprehensive theoretical argument that links the levels of party system institutionalization not only to the frequency of vote buying, but also with the citizens' decision to sell or nor their votes. Specifically, I suggest that high electoral volatility and weak parties engender high levels of uncertainty about the electoral outcome which, in turn, incentivize politicians to insure them against the possibility of electoral defeat by buying the votes of their constituents.

To begin with, weakly (fluid) institutionalized party systems are characterized by high levels of electoral volatility (Mainwaring and Scully 1995; Mainwaring 1999; Mainwaring and Torcal 2006; Moser 2001). When electoral volatility is high, elections are always held in changing environments, involving new issues, and high turnover of parties and party elites. In this dynamic environment, party labels are likely to mean little to most voters who usually lack of long-term commitments with determined parties and constantly change allegiances (Madrid 2005; Zielinski et al 2007).[50] Given that voters lack ideological attachments to a political party, the number of its core supporters shrinks, increasing the number of swing or marginal voters. An increasing and ever changing number of floating voters, in turn, generates important fluctuation in the change and composition of the electorate.

In addition, given that swing voters have no allegiance to any political party, all parties, small and big, from all points along the ideological spectrum approach floating voters knowing that they could make the difference between electoral defeat or victory, generating a great dispersion of the vote (Moser 1999, 2001) and, consequently, a substantial

[50] In other words, voters rarely make electoral decisions based on ideological and programmatic basis.

increase in the level competitiveness of the electoral race across the political spectrum (Lachat 2009).[51]

Together, a dynamic electorate and highly competitive elections, make it very difficult for politicians and parties to correctly predict the outcome of any given electoral race thus introducing a substantial level of uncertainty into the candidates' decision process. This uncertainty is exacerbated by the fact that political parties in fluid party systems generally have anemic organizations and limited resources to mobilize voters (Mainwaring and Scully 1995; Hicken 2002; Mainwaring and Torcal 2006), so they can do very little for their candidates on election day. As a result, candidates must do everything they can to bolster their own personal support network—even at the expense of their own party's reputation.[52] As Suchit (1990) emphasizes:

> "Parties' policies and performance were not a determinant of voting behavior, particularly in the provinces. Thus, in their election campaigns… politicians concentrated on their individual policies, achievements, and patrons, and rarely emphasized the party's performance. They had to set up their own election campaigns and campaign organizations, financing and recruiting their own staff…[Their] electoral success depended on their own efforts rather than [the efforts of] the party." Suchit (1990: 261)

Since, as suggested by Díaz-Cayeros et al (2007), politicians try to minimize the possibilities of electoral defeat, an increase in electoral uncertainty may effectively lure them to seek other ways of feathering their beds to insure themselves against possible political loss in the next election (Birch 2001: 2). In other words, an increase in the level of electoral

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[51] I am specifically referring to the level of "inter-party" competition.

[52] Results from a survey conducted during the 2008 local election in Nayarit, Mexico, suggest that voters very rarely remember the name of the candidate who offered money in exchange for votes (only 5% remembered the candidate's name). Yet, almost 80% of them could tell the name of the political party that such candidates were representing.

uncertainty triggers their desire to resort to vote buying as a profitable electoral strategy.[53] Because swing voters are unlikely to be attentive to the political cues emitted by political parties suggests, all else being equal, as the number of marginal votes that ensure electoral victory increases so does the costs of appealing to those voters using programmatic appeals. Hence, rather than spending on advertising political platforms, politicians aim at developing private and personal relationship with the electorate, and offer money for their votes. Wang and Kurzan (2007: 75) underscore this point nicely:

> "…some campaigns may keep costs down by targeting their vote buying efforts at swing voters, rather than wasting resources on ideologically or otherwise committed voters."

Although vote buying is not the only possible strategy when candidates campaign on personal vote getting strategies rather than on the reputation or political position of the parties, it seems to stand out as one of the preferred methods for most candidates.[54] Three reasons explain this point. First, campaigning on the basis of party accomplishments, however, could be very costly for politicians in developing democracies with fluid party systems since voters find it very difficult to credit political parties for particular outcomes. Second, as Hicken (2002, 2007) suggests, many developing countries traditionally have lax enforcement of anti-vote buying laws. This, in turn, not only creates a market for votes, but also increases the expected utility of this corrupt practice since politicians are unlikely to be severely punished if detected. Third, as mentioned before, vote buying reduces the

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[53] Indeed, as Szwarcberg (2010) argues, the uncertainty about electoral results induces risk-averse and career seeking candidates to employ clientelistic strategies to turnout working and low income voters.

[54] Candidates can, for instance, offer pork or rely on intimidation and violence. Yet, empirical evidence in Brazil (Samuels 2002) and Thailand (Ockey 1991; Hicken 2002) suggests that pork is not linked to electoral support thus making it a less attractive electoral strategy than vote buying. In addition, as Hicken (2002) reports, violence and intimidation must be conducted in a large enough scale to achieve the expected electoral results, making it a very costly strategy.

uncertainty associated with low levels of party system institutionalization, serving as a cost-effective insurance against possible electoral defeat.

A voluminous literature in economics and political science provides theoretical support for the swing voter logic described above, arguing that swing voters are more easily swayed away by monetary transfers because of their ideological indifference between opposing parties (e.g. Dixit and Londregan 1996; Persson and Tabelini 2000; Stokes 2005). In addition, Frohlich and Oppenheimer (1984) also suggest that politicians may find it optimal to target income transfers to the most ideologically moderate or swing voters. [55, 56] Empirical and anecdotal evidence in Argentina, Costa Rica and Mexico offer support for these claims.[57] In a survey conducted between 2001 and 2002 in three Argentine provinces, for example, Stokes (2005) finds that only a very small proportion of those who rated the Peronist Party as "very good" received some kind of material or monetary offer from that party.[58] The findings presented in Lehoucq and Molina (2002) also confirm that uncertainty in the electoral outcome led parties to experiment with vote buying. Specifically, they show that in a context of competitive elections and during the presidencies of many men who did not take sides in electoral combat, parties were left with little option but to stuff the ballot

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[55] A different strand of literature, pioneered by Cox and McCubbins (1986), argues that partie's core constituents are more reliably responsive to monetary transfers than swing voters. Cox (2007), Nichter (2009) and Dunning and Stokes (2008) further note that buying the votes of their constituents produces bigger electoral rewards since vote buying is an effective mobilizing strategy.

[56] As Cheng (2008: 5) cogently notes: "The difference between the core and swing voter models stem from a disagreement about the expected relative responsiveness of core and swing voters upon receiving distributive benefits.

[57] Chang (2004) also finds empirical support for this claim in Italy. Specifically, Using data from pre-1994 Italian legislative elections, Chang (2004) demonstrates that politician's electoral uncertainty regarding their prospects of winning reelection drives them to increase their propensity for corruption.

[58] For a detailed description of the survey methodology and exact questions, please see: Stokes, S. (2005). "Perverse Accountability: A Formal Model of Machine Politics with evidence from Argentina," American Review of Political Science 99(3): 315-25.

box or try to ply voters with money and favors (Lehoucq 2007: 40). Moreover, in a recent interview with a local newspaper in Oaxaca, Mexico, a PRI party broker stated:

> "The coalition (PAN-PRD) outplayed us with better electoral engineering… Money played a fundamental role in the electoral strategies. Members of the coalition had more money and swayed a lot of voters to vote for them offering money…"[59]

We still need a systematic theory linking weak party organizations and vote buying. Empirical evidence from developing countries indicates that candidates prefer vote buying over other electoral strategies when their parties are weak. For instance, Hicken (2002) finds that candidates in Thailand cultivate personal support networks by buying the votes of their constituencies. Similarly, Pfeiffer (2006) links the weakness of political parties with this practice. Moreover, in a well-crafted quantitative study of social transfers in Mexico, Díaz-Cayeros et al (2007) show that the Action National Party (PAN) used the most important antipoverty program (Oportunidades) to gain the electoral support of voters in municipalities in which the aforementioned political party has been traditionally weak. Relying on survey evidence from different developing countries, Transparency International, concludes that: "Vote buying tends to be carried out where parties are weak, with elections centered on candidates rather than parties, and where traditions of patronage are engrained…"[60] Thus these examples provide empirical evidence for the claim that vote buying is more likely to occur in countries where political parties are weak.

In sharp contrast to higher likelihood of vote buying in countries with weakly institutionalized party systems, I argue that this phenomenon is less frequent in nations

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[59] Extracted from: "Al Margen Oaxaca: Descifrando la Realidad." http://almargenoaxaca.wordpress.com/

[60] See Transparency International. 2004. "Global Corruption Report."

where parties and party systems are highly institutionalized. In stable party systems, political parties have strong roots in society and voters generally cast their vote on the basis ideological or programmatic basis (Downs 1957; Hinich and Munger 1994); and party organizations acquire independent status and value of their own (Huntington 1986). Moreover, as Przeworski (1975) and Mainwaring and Torcal (2006) suggest, the patterns of political competition manifest regularly in these systems, leading to considerable stability in the electoral market. Together, these characteristics of stable party systems have three effects on the politician's incentives to use vote buying a viable electoral strategy.

First, they reduce the number of votes that are available to purchase. Since ideology plays a fundamental role in influencing the voters' decision, the number of swing or marginal voters is arguably lower in these countries. Since voters have long-established ideological identities, it is unlikely that they will be swayed away with strategies that do not involve party platforms. Moreover, given that only few votes are available for purchase their price can be substantially higher, reducing their marginal utility. Second, the stability in the patterns of political competition reduces the politicians' uncertainty about electoral outcomes since candidates are likely to forecast the outcome of elections more easily. Third, countries with stable party systems are more likely to have other democratic institutions that efficiently enforce anti-vote buying laws by severely punishing candidates or parties that engage in this practice, thus substantially increasing the risks of offering money for votes and, consequently, its possible costs. This, in turn, implies that candidates will see vote buying as a less profitable activity, reducing its likelihood.

In sum, the arguments delineated above suggest that the level of party system institutionalization can fuel vote buying. This leads me to state the following hypothesis:

Hypothesis 2.1: The lower (higher) the level of party system institutionalization, the more (less) vote buying will occur.

Hypothesis 2.1, which is depicted graphically in Figure 1, puts forth a novel prediction about the effects that party systems institutionalization may have on the politicians' incentives to engage in vote buying in developing countries, but insufficiently explains an additional interesting puzzle: Does the level of party system institutionalization affect the citizens' vote selling behavior? I put forth an answer to this question in the following section.

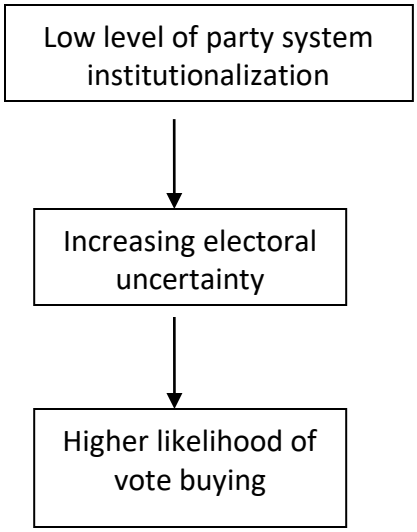

Figure 2.1: The Supply Side of Vote Buying, Causal Chain

2.3.2 Demand-Side of Vote Buying

In the previous section, I suggested that high electoral volatility and weak political parties, which are two key characteristics of fluid party systems, are likely to alter the

candidates' strategic calculations and thus their electoral behavior by introducing a high degree of uncertainty about the potential electoral outcome.[61] But the effect of uncertainty is, however, not limited to the strategic behavior of parties and candidates. Indeed, as Sartori (1986) and Moser (2001) suggest, electoral volatility also affect the citizens' voting behavior. This is because fluid party systems not only provide voters with very limited information cues about the election, but also with poorly developed sources of political information (Moser 2001). As Desposato (2007) puts it:

> "…many voters have only a rudimentary or even no formal education. In addition, many citizens have very limited access to the media and other forms of political information."

Since voters in these systems lack good sources of political information, it is not surprising that they are often uncertain about the utility of voting and unaware that an activity such as vote buying is harmful for democracy and potentially pervasive.[62] For example, a survey conducted during the 2008 local elections in Nayarit, Mexico, the Mexican state from which I present evidence, suggests that voters do not see vote buying as a pervasive phenomenon. Specifically, voters were asked: "How pervasive do you think vote buying is in Mexico?" In all, 55% responded that vote buying is a severe problem in Nayarit, 23% reported that vote buying is not a problem and the remaining 22% did not know. The lack of knowledge about the potential negative consequences of vote buying is not limited to Mexico. In Nigeria, where vote buying also infects elections, Bratton (2008) documents that

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[61] Specifically, I suggested that high levels of electoral volatility and weak parties provide politicians with strong incentives to resort to vote buying as a profitable electoral strategy.

[62] Vote buying is a political decision as many others that voters make during an electoral process.

7% of Nigerians think that vote buying is not wrong at all.[63] But the numbers are not any better when one looks at the perceptions of people about vote buying. As Bratton (2008) notes:

> "… Nigerians are less critical of their own behavior. Barely half of them (49%) think that it is "wrong and punishable" for a voter to accept money in return for a vote. The other half of the adult population is willing to excuse participation in a vote buying transaction as "wrong but understandable" (35%) or not "wrong at all (10%)." (Bratton 2008: 623).

The Mexican and Nigerian cases are interesting but, as mentioned before, arguably not surprising. Indeed, two reasons explain why voters are often unaware or indifferent about the existence of vote buying in fluid party systems. First, the transaction costs of obtaining information about vote buying are very high. Voters need a substantial amount of time and effort to obtain information, assimilate and incorporate it into their decision-making process. In addition, voters generally have to make very rapid decisions when deciding whether or not to accept a monetary offer from political parties or candidates. But when voters lack developed sources of information their decision is surrounded by a lot of real and subjective uncertainty.[64] Hence, voters often make the decision about engaging in vote buying based on probabilistic predictions (guesses) about the costs and benefits of engaging in it. Since the transaction costs of obtaining information and uncertainty about their decision are high, the citizens' incentives to obtain information about vote buying are low.

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[63] Data drawn from the pre-election Afrobarometer survey in Nigeria (2007).

[64] As mentioned before, sources of political information in weakly institutionalized party systems are likely to be underdeveloped.

Second, vote buying is illegal, prohibited in almost every democracy around the world. Its criminal nature prevents parties, candidates and even voters from revealing its existence. The lack of information about the potential consequences of this phenomenon ironically creates space for it because makes politicians are less accountable to their constituencies. Moreover, given that voters rarely have trustworthy information about the frequency and consequences of this instance of electoral corruption, they may simply assume that it does not exist or, simply, make them more likely to sell their votes.

Based on the preceding discussion, it thus seems plausible that voters in countries with fluid party systems consider other factors rather than income or reciprocity (e.g. how widespread the vote buying phenomenon is, and whether or not selling their votes has costly consequences) when deciding whether to sell their votes.[65] In what follows, I develop a simple causal story that explains how and why the lack of political information, which is common in fluid party systems, influences citizens' attitudes towards vote selling and their decision to exchange votes for money. The argument suggests that the citizens' decision to sell their votes during elections is based on the available information about the phenomenon of vote buying, even if the amount of information collected is limited. More specifically, as the voter's exposure to information about it increases, so does her/his ability to gather and understand the effects of the monetary offers, and this, in turn, reduces the likelihood to exchange votes for money.

Existing studies of opinion formation have already documented that most citizens do not have very stable or "real" attitudes toward even the major political issues of

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[65] See Brusco et al (2004); Stokes (2005); Nichter (2009); Finan and Schechter (2009); Magaloni (2006).

the day (Converse 1964, 1975; Zaller 1992).[66] This is because voters rarely have the necessary information that helps them to form and maintain attitudes with respect to important political events. As Zaller (1992) notes:

> "The impact of people's value predispositions always depends on whether citizens poses the contextual information needed to translate their values into support for particular candidates and the possession of such information can… never be taken for granted." (Zaller 1992:25).

Moreover, political psychologists such as Krosnick (1998), Lavine et al (1996) and Alvarez and Brehm (2002) have shown that information plays an important role in forming political attitudes and consequently in affecting the political behavior of voters.

The aforementioned studies suggest that information has two main effects on the formation of political attitudes and their subsequent effect on voter behavior. First, information reduces attitude uncertainty. As the information received by individuals increases, so do their certainty about their attitudes and the stability of such attitudes over time. Second, information makes predispositions and values relevant for beliefs about specific issues (Binzer-Hobolt 2005:17). In other words, information flows allow citizens to link attitudes with specific issues, enabling them to act on the bases of these attitudes. In the context of vote buying, the argument discussed above provides with a framework for understanding the voters' attitudes towards vote buying and their predispositions to be swayed away by corrupt politicians in fluid party systems. Specifically, the lack of developed

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[66] These effects have been extensively documented in developed nations (See, Converse (1964, 1975) and Zaller (1992).

sources of political information may affect the citizens' predisposition towards vote buying in two different ways.

First, lack of information about this political phenomenon tends to minimize the salience of vote buying in the public debate. Despite its frequency, vote buying does not occupy a central stage in the political debate. This is because, as mentioned earlier, neither politicians nor voters have incentives to reveal its existence. Information about vote buying provides citizens with an environment that fosters dialogue about the existence and ubiquity of this phenomenon. In its absence, the dialogue among citizens about this phenomenon decreases, creating an environment that diminishes the citizens' general interest and understanding about its frequency and consequences. This, as suggested by the existing literature on political behavior, disincentives citizens to invest time and effort in making political judgments (Kuklinski et al 2001; Le Duc 2000; Binzer-Hobolt 2005). As a result, by diminishing salience, lack of information does not allow voters to better understand the frequency and severity of vote buying.

Second, limited political information increases the citizens' uncertainty not only about the frequency of vote buying, but also regarding the candidates' promises. The criminal nature of vote buying and the possibility of being punished by the electoral authorities induce political parties and candidates to buy votes surreptitiously. This not only makes it difficult to document politicians' blatant attempts to manipulate elections buying votes, but also prevents citizens from obtaining useful information about the frequency and severity of these practices. When citizens lack information about vote buying, they may not know how frequently it occurs, or how severe it is. That is, the lack of credible information increases the voters' uncertainty about the real frequency and severity of vote buying. In addition, the lack of information generates a great deal of uncertainty about the future

benefits of voting based on ideological or programmatic appeals so they rapidly discount promises of future benefits making vote buying attractive to them (Hicken 2002: 17). Following these lines of reasoning, I thus state the following hypothesis:

Hypothesis 2.2: The lack of political information, which is a common feature in weakly institutionalized party systems, decreases the likelihood that voters will perceive that vote buying is high in these systems.

Hypothesis 2.2 predicts the effects that information flows may have on the citizens' perception of the frequency and severity vote buying, but insufficiently explains an additional interesting puzzle: Does information about its negative consequences affect the citizens' vote buying behavior? Lack of political information, I suggest, makes citizens more likely to engage in the corrupt electoral practice of vote buying.

On the one hand, citizens are unlikely to develop a set of rules against it. In the absence of information about the potential consequences of vote buying, it is virtually impossible for voters to understand the consequences of selling their votes. So it is difficult for voters to learn not only that vote buying is pervasive, but also that the act of selling votes by individuals may sum-up in the aggregate to have negative effects on democracy and democratic consolidation, violating their own political future and independence and affecting the political and personal rights of other members of the society. Given that voters cannot recognize that vote buying not only affects their own personal rights but also democracy in the aggregate, they cannot develop a revulsion toward vote selling. Conversely, when voters are provided with any piece of information about this practice, they are more likely to learn that the frequency of vote buying is high and that it can indeed have negative consequences in the aggregate.

On another, the lack of information about the potential consequences of vote buying prevents voters from learning and internalizing norms against it. This, in turn, impedes voters to process and understand why rejecting a monetary offer from a political party and/or candidate is of value and makes sense, until finally they accept them as their own view point. Put differently, information about the negative consequences of vote buying makes it more difficult for individuals to internalize the set of rules (i.e. norms) against this phenomenon. By contrast, more information about this practice makes it easier for citizens to learn and internalize norms against vote buying, thus making them more likely to accept money in exchange for votes. Once the norms against vote buying are internalized, their behavior is likely to reflect those norms, making them less likely to engage in vote buying.

Although the empirical evidence supporting the claims posited above is still scarce in the realm of comparative politics, two studies have indirectly documented the effects that information flows may have on clientelism and vote buying. For instance, Vicente (2007) demonstrated that directed voter education may be highly effective in curtailing vote buying in Sao Tome Principe.[67] In a fascinating study, Wantchekon (2003) showed that information influences voting behavior in Benin. Wantcheckon's results are cogently summarized in the following quote:

> "…The results also indicate a significant and positive treatment effect for all candidates. In addition, there is a significant and negative public policy treatment effect for Northern candidates. By contrast, there is a positive treatment effect for Southern candidates." (Wantcheckon 2003: 413-414).

[67] Both Wantchekon (2003) and Vicente (2007) use randomized field experiments to test their hypotheses.

Besides the experimental evidence mentioned above, the idea that information about vote buying affects the voters' predisposition to engage in vote buying can also be illustrated using qualitative evidence. Indeed, in an interview conducted during the 2008 local elections in Nayarit, Mexico,[68] a grassroots party organizer stated:

> "…people (voters) accept money in exchange for votes because they lack information about vote buying. Yes, they don't really know anything about it. Worse, they do not have time to think about the pros and cons of selling their votes. We –the political party-, obviously, have no incentives to reveal the reality behind vote buying because they will learn that it is a corrupt practice and, perhaps, get mad at us, punish our candidate the day of the elections or simply reject the money."

The preceding discussion thus allows me to state the following individual-level hypothesis:

Hypothesis 2.3: The lack of political information, which is a common feature of fluid party systems, increases the likelihood that voters will engage in this practice.

If the level of institutionalization of the party systems and the quality of the information flows that exist in these polities affect the micro-level strategic vote buying (selling) behavior of both politicians and voters in such a way that they can explain why vote buying is more likely to occur in some countries than in others, a more interesting question arises. Namely, to what extent does party system institutionalization affect an aggregate political outcome such as electoral fraud? I put forth an answer to this question in the following section.

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[68] Interview conducted by the author in Huajicori, Mexico.

2.4 Party System Institutionalization and Electoral Fraud: The Macro-Level

In the previous section, I developed a theoretical argument linking the level of institutionalization of the party systems to the likelihood of vote buying. But, does the micro-level vote buying behavior have any consequences for aggregate political outcomes? In what follows, I explore whether and how the level of party system institutionalization in developing countries affects the likelihood of electoral fraud.[69]

That election fraud overshadows elections in both developed and developing countries is a well-known fact. For instance, as Alvarez et al (2008) document, in 2006 alone, the presidential elections in Mexico, Italy and several former Soviet republics were surrounded by significant allegations of electoral corruption (i.e. fraud). The frequency and salience of this modality of political corruption has motivated political scientists to formulate explanations for its existence. While some existing studies have found that electoral fraud is correlated some socio-economic factors (Eisendstadt 1999Trejo and Aguilar-Rivera 2002; Lehoucq and Molina 2002; Alston and Gallo 2009; Ziblat 2009), others have suggested that some of the observed variation in electoral fraud can best be explained by political competition (Simpser 2005),[70] or other institutional factors such as electoral rules (Lehoucq and Molina 2002). While these factors may encourage politicians to rig elections in developing countries, they pay little attention to the micro-foundations of electoral

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[69] Scholars such as Mainwaring and Scully (1995), Stoner Weiss (2001), and Mainwaring and Torcal (2006) have suggested that fluid arty systems can have negative effects on representative democracy. Yet, the link between party systems and electoral corruption has neither been theorized nor systematically tested.

[70] With some important exceptions, Simpser (2005) finds a positive correlation between competitiveness, measured as the percentage difference in votes between the first and second place in the election, and various measures of electoral fraud.

corruption.[71] By contrast to existing explanations of electoral fraud, I argue below that by increasing the likelihood of vote buying at the individual level, weakly institutionalized party systems trigger the incumbents' desire to use electoral corruption to hold their offices.[72]

In the previous section I suggested that the unstable patterns of electoral competition together with the important lacunas of political information that are common in fluid party systems affect politicians' and voters' strategic behavior in such a way that they increase the size of the group of "swing" voters, making vote buying a potentially profitable electoral strategy. Although the existing theoretical literature on vote buying emphasizes that vote buying is generally practiced by incumbents, the empirical and anecdotal evidence suggests otherwise.[73] For instance, an original survey conducted across different municipalities in Nayarit, Mexico,[74] reveals that more than 50% of the monetary offers received by voters in the 2008 local elections came from opposition parties. A report from The Philippine Daily Inquirer also suggests that both incumbents and opposition bribed voters in the 2010 local elections in the Philippines.[75] As a voter in Abasolo, Mexico said:[76]

> "…those times when the PRI was the only party offering construction materials, food, or money in exchange for votes are long gone. Well, please don't get me wrong but I don't regret it at all, it is better now! When election time approaches candidates and party

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[71] I use the terms electoral corruption and electoral fraud indistinctively.

[72] I assume that survival and longer time-horizons in office maximize the incumbents' utility functions.

[73] Magaloni et al (2006); Stokes (2007); Nichter (2008).

[74] Original survey conducted by the author in the 2008 local elections in Nayarit, Mexico.

75 http://newsinfo.inquirer.net/breakingnews/nation/view/20100812-286350/Less-cheating-vote-buying-in-May-polls-says-Pulse-Asia-survey.

76 Interview conducted by the author in May 2009, before the 2009 mid-term federal elections in Mexico.

> operatives from the PAN, PRI and PRD suddenly appear bringing all sorts of goodies. Of course, they ask for our votes before delivering the goods, but not all people vote for them"[77]

The fact that parties that vote buying is practiced not only by incumbent parties and candidates but also from the opposition has, in turn, one fundamental consequence on the likelihood of electoral fraud. Vote buying is an individual-level phenomenon in which both candidates and voters participate. Given its illegal nature, proving its existence is challenging as neither politicians nor citizens have incentives to truthfully cooperate against organizers of such schemes. Therefore, allegations of vote buying are difficult to prosecute and punish (Alvarez et al 2008) and candidates from all parties face increasing incentives to buy votes.

By resorting to vote buying, opposition parties and candidates not only increase the level of competitiveness of the election but also have credible chances to flip the outcome of the election in their favor. This is because, as mentioned before, the market of votes in countries with weakly institutionalized party systems is likely to have a majority of floating voters who can effectively be swayed away with money or other material goods. Therefore, the widespread use of vote buying increases the uncertainty about the incumbent's prospects of survival in office. Since retaining office maximizes the incumbents' utility function,[78] an increase in electoral uncertainty triggers a corrupt response from the incumbent government which, in turn, increases the likelihood of electoral fraud.

More specifically, an increase in electoral uncertainty can trigger at least two types of corrupt electoral behavior from incumbent governments. On one hand, incumbents can also

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77 Other citizens in different municipalities and states in Mexico backed up this statement. As in this case, the interviews were conducted by the author.

78 I assumed before that politicians are utility maximizers who seek to win elections in order to extend their time horizons in office.

use vote buying as a means to secure an electoral victory and play a "blame-game" in which the opposition can also be easily accused of corrupting elections. The pervasive use of vote buying at the individual-level can thus sum-up in the aggregate, increasing the probability of electoral fraud. On another, incumbent governments can also respond to an increasing electoral uncertainty by manipulating the election secretly placing fictitious names on the voter rolls, casting bogus votes in the names of persons who did not vote, obtaining and marking absentee ballots without the active input of voters involved, and falsifying vote tallies. In other words, they can stuff the ballot box. Lehoucq and Molina (2002: 241) underscore this as follows:

> "The possibility of gaining advantage through underhanded maneuvers was enough to unsettle most parties and to increase political uncertainty…Regardless of what parties knew or believed, it was rational to charge opponents with stuffing the ballot box…"

The study of the causes and consequences of electoral fraud is still in its infancy, but several studies have already suggested that political factors, especially political competition, explain some of the observed variation.[79] For instance, using evidence from Germany in the 19th century, Ziblatt (2009) finds that election fraud occurred more frequently in districts with high level of electoral competitiveness. Lehoucq and Molina (2002), and Eisenstadt (1998, 1999) also suggest that political competition heavily influenced the likelihood of electoral fraud in Costa Rica and Mexico respectively.[80] Moreover, in the most recent

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79 Simpser (2004) suggests that election fraud also occurs in many noncompetitive elections, arguing that it can be means by which autocrats demonstrate their dominance of the system.

80 See Lehoucq and Molina (2002) for Costa Rica and Eisenstadt (1998, 1999) for more details on the Mexican case.

gubernatorial elections in Puebla and Veracruz, Mexico, both surrounded by a high degree of electoral uncertainty, the incumbent governments were accused of rigging the election in favor of their candidates.[81] Thus, it seems plausible to suggest that weakly institutionalized party systems and the electoral competition that emerges from them provide incumbents with important incentives to resort to electoral fraud in order to maximize their chances of political survival and extend their time-horizons in office. This leads me to state the following hypothesis:

Hypothesis 2.4: The lower (higher) the level of party system institutionalization, the higher (lower) the likelihood of electoral fraud.

2.5 Concluding Remarks

The theoretical argument developed in this chapter allows us to better understand how and why vote buying is more common in some countries than others. By contrast to existing studies on the determinants of vote buying (Hicken 2002; Brusco et al 2004; Schaffer 2006; Stokes 2007; Nichter 2008; Nicther et al 2010; Finnan and Schechter 2009), that explain vote buying on the grounds of socio-economic factors and reciprocity, the theory presented in this chapter pinpoints the micro-level effects of party systems –an important political institution- on the likelihood of buying and selling votes in new and developing democracies.

Stated briefly, I argued that the unstable patterns of interparty competition that generally exist in weakly institutionalized party systems not only increase the number of "swing" or "floating" voters thus generating a profitable market of votes, but also

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81 The PRI (incumbent) won the election in Veracruz but not in Puebla.

substantially increase electoral uncertainty among competing candidates regarding the outcome of the election. Given that programmatic or ideological cues rarely influence the swing voter's decision, candidates resort to vote buying trying to insure themselves against possible electoral defeat. Conversely, the stability in electoral patterns of political competition in highly institutionalized party systems and the absence of a surplus of marginal votes deters politicians in these countries from engaging in this act of political corruption. Therefore, vote buying will be more common in countries where the party system is weak than in those with a highly institutionalized party system.

I also suggest in this chapter that the lack of political information in weakly institutionalized party systems has two consequences on voter behavior. First, the lack of information about the consequences of vote buying tends to reduce its salience in the public debate and increases the uncertainty about its frequency, preventing voters to understand more clearly the frequency and severity of vote buying. Furthermore, since voters cannot understand that vote buying is a severe problem, they find it difficult to develop and internalize norms against it thus increasing their likelihood to engage in vote buying.

The theoretical framework developed in this chapter has two important implications. First, several scholars have already suggested that poverty, constituency size, reciprocity, or intra-party competition fuel vote buying (Cornelius 2004; Stokes 2005; Magaloni 2006; Brusco et al 2004; Finan and Schechter 2009). By contrast, the arguments developed in this chapter suggest that an institutional variable such as the level of party system institutionalization can explain why vote buying occurs more frequently in some countries or regions than in others. Since, as mentioned before, this phenomenon varies substantially across different political units the theory presented in this chapter thus provides us with a much more comprehensive explanation of this act of political corruption. Second, this

chapter also deepens our understanding of the political consequences of this phenomenon by explaining how and why vote buying increases the likelihood of aggregate electoral fraud. Thus, it presents a tangible explanation of the potential negative consequences of vote buying on democratic survival and consolidation.

Put together, the theoretical argument led me to state the following hypotheses:

Hypothesis 2.1: The lower (higher) the level of party system institutionalization, the more (less) vote buying will occur.

Hypothesis 2.2: The lack of political information, which is a common feature in weakly institutionalized party systems, decreases the likelihood that voters will perceive that vote buying is high in these systems.

Hypothesis 2.3: The lack of political information, which is a common feature of fluid party systems, increases the likelihood that voters will engage in this practice.

Hypothesis 2.4: The lower (higher) the level of party system institutionalization, the higher (lower) the likelihood of electoral fraud.

In the next three chapters, I systematically test the hypotheses and causal mechanisms underlying them using a multi-methodology approach that include comparative case-study research, a field experiment, cross-national survey data, statistical tests conducted on a large-n pooled dataset, and statistical evaluation of within-country data on vote buying and party system institutionalization for some countries.

CHAPTER 3:

PARTY SYSTEM INSTITUTIONALIZATION AND VOTE BUYING: THE QUANTITATIVE EVIDENCE

3.1 Introduction

In the previous chapter, I suggested that the level of institutionalization of the party system explains why vote buying is more frequent in some political units than in others. Specifically, I theorize that candidates and/or political parties in party systems with low levels of institutionalization are more likely to use vote buying as a latent electoral strategy than those in systems with high levels of institutionalization (H 2.1).[82] I also theorized that the lack of political information about vote buying has two important consequences on voter behavior. First, by boosting the salience of vote buying in the public debate and reducing uncertainty about its pervasiveness and frequency, information about the negative consequences of this illegal electoral strategy creates a *learning effect* that helps citizens to understand the frequency and severity of vote buying (H 2.2). Second, the learning effect that results from receiving information about its negative consequences allows citizens to develop and internalize *norms against vote buying*, making them less likely to exchange votes for money (H 2.3).

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82 This is because low levels of party system institutionalization introduce a high degree of electoral uncertainty regarding the electoral outcome. Since vote buying can be a profitable strategy, they resort to this corrupt practice in order to secure the electoral victory.

In this chapter, I statistically test these hypotheses using two different methodologies. To test hypothesis 2.1, I use data from an original survey on vote buying and clientelism gathered in the 2009 mid-term legislative elections in Mexico. In addition, I test hypotheses 2.3 and 2.4 using an original randomized field experiment conducted during the 2008 local elections in Nayarit, Mexico. Regarding hypotheses 2.3 and 2.4, the results of a series of hierarchical logit models indicate that electoral volatility has a substantial and highly significant statistical effect on the likelihood of vote buying thereby supporting my theoretical expectations. Likewise, statistical results suggest that individuals in the treatment group not only perceive this instance of electoral corruption as a very frequent phenomenon, but also are less likely to engage in vote buying.

The quantitative results of this chapter have important implications for the existing literature on the causes of vote buying, and parties and party systems. For example, the empirical tests presented here indicate that institutional factors (i.e. the level of party system institutionalization) have substantive and highly significant statistical effects on the likelihood of vote buying. The strong statistical support for the first three hypothesis of my theoretical argument (H 2.1, H 2.2, and H 2.3) suggests that scholars need to pay more attention to the institutional factors that engender this instance of political corruption. My results also suggest that the structure of party systems can have important consequences for democracy in developing nations. By increasing the likelihood of vote buying, inchoate party systems are likely to endanger the survival of democracy in these nations.

The rest of the chapter is organized as follows. In the next section, I carefully describe the sample, statistical methodology, dependent variable, independent and control variables used to test hypothesis 2.1, and present the results from testing this hypothesis. I then describe the randomized field experiment, data, variables and statistical methods, and

robustness tests used to test the statistical validity of hypotheses 2.2 and 2.3. I end with a brief summary of this chapter and discuss its implications.

3.2 Testing Hypothesis 2.1

3.2.1 Sample

I test the effects of party system institutionalization on my dependent variable (i.e. vote buying) using the answers of an original survey on clientelism and vote buying conducted immediately after the 2009 mid-term legislative elections in Mexico. The national survey was administered from July-August 2009 in 100 electoral districts in Mexico using the probability proportional to size systematic method. Using the typology of the sections pre-defined by IFE as: Urban, Rural and Mixed (Urban/Rural), we first determine the distribution of the population within each type. We then assigned the number of cases due to this distribution. We then proceed to obtain the complex stratified sample, with a probability proportional to size systematic (PPS Systematic) method. This method requires the definition of another variable, measure of size (MoS) of each electoral section in order to calculate the probability of selection for each section. The stratification variable is the typology, so we then proceed to specify the number of units to sample per stratum (this is according to the distribution of the stratification variable).

Then we proceed with the second stage in the sampling method. Since we need to guarantee national representativeness, we cluster by state ID which we have previously distributed by stratification variable. Each electoral section has an address that refers to the place where the polling booth is installed. If this address is a household the surveyors start their interviews on it, if not they go to their right, discard three houses and attempt to get a

survey in this house. This system continues, they follow a spiral shaped pattern always to the right. Usually, each surveyor makes 10 surveys per section. The survey was carried out by SIMO, a private firm with ample experience in survey research in Mexico. The questionnaire was made available in Spanish. Figure 3.1 describes the sampling strategy.

Stud	Door – to - door survey
Survey	July 24 -27,
Target	
Sampling	IFE's nationwide electoral
Sampling	Probability Proportional to Size Svstematic method
Household	Simple systematic method
Observation units selection methods	Simple random sampling
Survey	Face – to - face

Representativene	National
Sample	1000 cases + 250
Special	Rural Oversampling (250
Sampling	(+/-) 2.7
Confidence	95 %
Response rate	1 out of 3
Editing	
Survey	30 minutes

Figure 3.2: Sampling Strategy.

The target population of the survey was that of people with 18 years of age or older, who were able to vote according to the electoral rules in Mexico.[83] The survey interviews were supplemented by open-ended interviews with the voters. The sample size is fairly representative in terms of districts, geographic distribution, age, sex, and socio-economic status. The final sample size realized was 1250. The fact that my sample is fairly

83 In Mexico, people who are 18 years or older are able to vote given that they are registered in the national nominal list of voters (Instituto Federal Electoral, Mexico).

representative of the political, economic, and social characteristics of Mexico is advantageous because it allows me to generalize my findings if my hypothesis in confirmed by the quantitative evidence. I now briefly describe how the dependent variable is operationalized and then discuss the operationalization of the independent and control variables.

3.2.2 Dependent Variable

The dependent variable in this analysis, *Vote Buying*, is measured at the individual level based on responses to a question about the distribution of money in exchange for votes during the 2009 mid-term legislative elections in Mexico. Since I am primarily interested in analyzing the effects of electoral volatility on vote buying, I operationalize the variable labeled *vote buying* using the responses to the following question: During the most recent electoral campaign, did you see candidates or brokers from political parties offering money in exchange for votes? The possible answers to this question are: 1) Yes; 2) No, and 3) Don't Know/No answer. Responses to this questions are thus coded as a dummy dependent variable where y=1 indicates that the respondent witnessed monetary offers before the 2009 election and zero otherwise. In all, 26% of the respondents reported that political parties and/or candidates offered money in exchange for votes in the election, while 71% did not report vote buying. The remainder, 3%, did not know or did not answer the question.

The aforementioned question allows me to measure the voters' exposure to vote buying offers, but it has some limitations. For instance, respondents did not report whether they received and/or accepted a monetary offer from parties or candidates.[84] The data also contains no information about the magnitude of the gift. Finally, the answers do not provide

84 Gonzalez-Ocantos et al (2010) suggest, in addition, that social desirability bias may affect the inferences obtained when respondents are asked about a sensitive/illegal issue such as vote buying. This is, of course, an important objection.

any information about the timing of the vote buying offer. Despite these limitations, the lack of sources of information about vote buying across and within countries does not allow me to measure my dependent variable in a different way.[85] The fact that my sample includes responses gathered in the 32 states of Mexico introduces an important degree of variation across and within Mexican states and electoral districts, thus making me confident about the accuracy of my results. Figure 3.1 illustrates this variation at the state level.

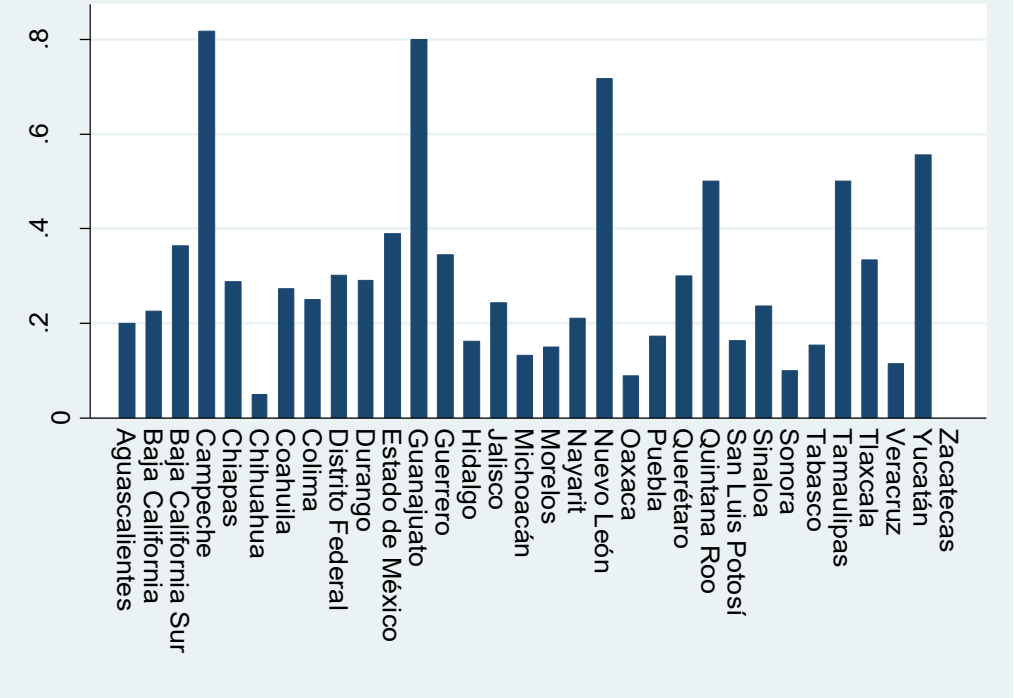


Figure 3.3: Vote Buying in Mexico, 2009.

3.2.3 Independent and Control Variables

Hypothesis 2.1 predicts a negative relationship between the level of party system institutionalization and the dependent variable -*vote buying*-. Given the structure of my

[85] The dependent variable used in this study resembles those used in previous studies of vote buying (Brusco et al 2004; Stokes 2005; Nichter 2008; Vicente 2008).

dataset, I thus need to measure the level of party system institutionalization for each electoral district in the sample. Political scientists have long debated about the characteristics and measures of party system institutionalization in developing countries (Mainwaring and Scully 1995; Coppedge 1997; Mainwaring 1999; Mainwaring and Torcal 2005). These scholars have measured the strength/weakness of party systems in less-developed countries using three characteristics: electoral volatility, the lack of ideological voting and personalism (Mainwaring and Torcal 2005). Given the limited availability of cross-national and cross-temporal data on either ideological voting or personalism across Mexican electoral districts, I use electoral volatility as my main measure of party system institutionalization.

The variable *electoral volatility* allows me to measure the system-level change in party vote percentage from one election to another (Roberts and Wibbels 1999: 580) and is operationalized using the Pedersen index. The Pedersen index of electoral volatility is calculated by summing the absolute value of changes in the percentage of votes gained or lost by each party from one election to the next divided by two.[86] The resultant index yields a scale between 0 (no parties gained, and thus no parties lost either) and 100 (all the parties from the last election were reduced to zero votes), because for every gain there is an equal (in terms of percentage of votes) loss. In other words, the 0-100 index is equal to the net percentage of voters who changed their votes.

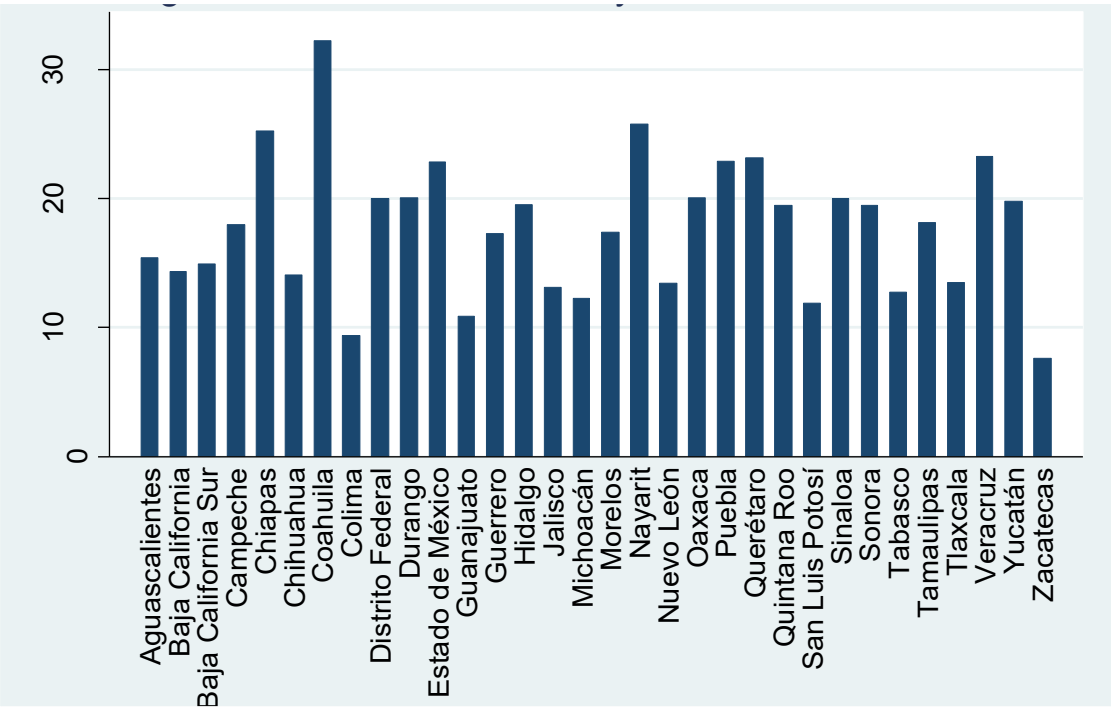
86 Dividing by two effectively ensures that the gains and losses are not effectively double counted (Mainwaring et al, 2009).

Following scholars such as Pedersen (1983), Roberts and Wibbels (1999) and Madrid (2005), I thus calculate the independent variable –*electoral volatility*- for the each electoral district in the 2009 legislative election in Mexico using the following formula:[87]

$$Volatility(V_t) = \frac{\sum_{i=1}^{n} |p_{i,t}|}{2}$$

$$0 \leq V_t \leq 100$$

The illustration presented in Figure 3.2 suggests that the main independent variable – i.e. *electoral volatility*- varies substantially across Mexican states and electoral districts. Figure 3.3 illustrates this variation. The mean of electoral volatility across the 100 Mexican electoral districts included in this study is 18.87 with a standard deviation of 6.89.[88]



[87] Data to measure this variable was drawn from: http://www. ife.org.mx. Calculations are available from the author upon request.

[88] The minimum value of electoral volatility is 5.94 and the maximum is 43.04.

Figure 3.4: Electoral Volatility across Mexican States, 2009.

Following existing studies on vote buying, I control for several socio-demographic, economic, and political variables. Several scholars have suggested that the incentives for vote buying should increase as the share of the population that is poor increases (Stokes 2007; Magaloni 2006; Kitschelt 2000; Gomez Tagle 2001). To this end, respondents in the survey were asked to select the range which best represents their household's monthly income. Answers to this question range from 1 (0 to 1299 MX Pesos) to 4 (8,000 or more MX Pesos). Hence, the variable *Income* is an ordinal variable ranging from 1 to 4. In line with existing studies, I expect a negative relationship between *income* and the dependent variable.

I also control for a series of socio-demographic control variables at the individual levels: *Age*, *Gender*, *Education, and Rural*. *Gender* is a dummy variable coded as 1 if the respondent is a male and zero otherwise. The individuals in the sample were also asked to select the category that best represents their educational level. The possible choices ranged from 1 (Illiterate) to 12 (Graduate degree). Therefore, the variable *Education* ranges from 1 to 12. Data for the variables *Age*, *Gender*, and *Education* is from the post-electoral survey described above. *Rural* measures the percentage of voters that inhabit in rural communities in each electoral district.[89] Data for this variable was drawn from the Mexican National Council of Population.[90]

I also use a political control variable to determine whether the respondents are core or marginal supporters of a determined political party: *Party ID*. Data for the ordinal variable *Party Ideology* is drawn from the aforementioned national survey. I coded *Party ID* in two

[89] Aparicio (2002) finds that vote buying is more frequent in rural than urban areas in Mexico.

[90] http://www.conapo.gov.mx

steps. First, the survey respondents were asked the following question: "Generally, would you consider yourself 1) Very panista, 2) Somewhat panista, 3) Very priista, 4) Somewhat priista, 5) Very perredista, 6) Somewhat perredista, 7) None, 8) N/A?" I then coded the ordinal variable *Party ID* as follows. *Party ID* takes the value of 1 if the individuals selected option number 7 (i.e. none). It takes the value of 2 is the respondents responded: somewhat panista, somewhat priista, or somewhat perredista. Finally, Party ID takes the value of 3 if the individuals identify themselves as very panista, very priista, or very perredista. I expect a negative relationship between *Party ID* and my dependent variable *vote buying*.

Finally, I control for the alternative hypothesis linking reciprocity with vote buying (Finan and Schechter 2009). To operationalize this variable, the survey asked respondents the following question: In your opinion: people who receive monetary offers from political parties and/or candidates feel the obligation to vote for those who offered money? Responses for this question range from 1 (very much) to 4 (no obligation at all). The variable derived from these responses is denoted as *Reciprocity*. I anticipate that the sign of *Reciprocity* will be negative.

3.2.4 Statistical Methodology

The original survey employed to test hypothesis 2.1 provides me with valuable information about vote buying and several political, economic, and socio-demographic indicators at the individual level. Yet it doesn't allow me to operationalize the main independent variable of this hypothesis (i.e. electoral volatility). The fact that, as described previously, electoral volatility is measured at the district level presents me with an estimation challenge since my main dependent and independent variable are measured at two different levels of analysis. To minimize the potential econometric problems that could arise given the

structure of my dataset, I estimate a series of multilevel logit random models (Judge et al 1985; Williams 1986; Greene 1990; Iversen 1991; Bryk and Raudenbush 1992).

Generally speaking, multilevel models explain micro-level outcomes in two ways: (i) by showing that parameters of models specified at the micro-level –where micro-level covariates are used to explain micro-level outcomes- are a function of context, and (ii) by showing that this micro-macro relationship can be expressed in terms of characteristics of the context, which takes the form of macro-level variables. In this analysis, the selection of a random effects multilevel models allow me to decompose the variance in the dependent variable into the within-context variance and the between context variance, and to study these two sources of variation for the micro-level outcome.[91]

3.2.5 Results and Analysis

Models 1-2 in table 3.2 report the results of the hierarchical logit models testing the effects of electoral volatility on vote buying. Specifically, in model 1, I report the results of a baseline model that includes the main independent variable, *electoral volatility*, and the following control variables: *Income*, *Gender,* and *Party ID*. The effect of *electoral volatility* in model 1 is positive and statistically significant at the 1% level in the multilevel logit specification which is estimated with random effects. In model 2, table 3.2, I added the control variables *Education*, *Age*, and *Rural* to the baseline specification of model 1. The results from model 2 suggest that the estimated coefficient on electoral volatility remains positive and statistically significant at the 1% level.

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[91] Random coefficient multilevel models are a type of variance components model.

In addition to electoral volatility, *Party ID* has the expected negative sign and is highly significant (1% confidence level). The negative and statistically significant relationship between this variable and vote buying suggests that the likelihood of vote buying decreases as the level of party identification increases. The coefficient on *Age* is also negative and statistically significant at the 1% level. This finding indicates that, holding other variables constant, parties or candidates are more likely to offer money to younger individuals.[92] Likewise, I do find statistical support for the alternative hypothesis that suggests that vote buyers are more likely to offer money to those voters who are more likely to reciprocate them with votes (Brusco et al 2004; Finan and Schechter).[93]

My results also indicate that there is no statistical relationship between poverty (i.e. socio-economic status) and my main dependent variable. The lack of statistical significance does not allow me to support the alternative hypothesis suggesting that political parties and candidates primarily target poor voters (Kitschelt 2000; Gomez Tagle 2001; Magaloni 2006).

TABLE 3.1

MULTI-LEVEL LOGIT ESTIMATIONS

[92] I lack space here to provide explanations about the causes of these relationships, but it is important to analyze in further detail in the future. In particular, the result that links Party ID with vote buying may provide insightful explanations for the core vs. swing voter debate.

[93] The coefficient on Reciprocity is negative and statistically significant at the 1% level.

	Model 1 DV: Frequency of Vote Buying	*Model 2 DV: Frequency of Vote Buying*
Electoral Volatility	.031***	.028***
	(.012)	(.013)
Income	.279	.331*
	(.183)	(.188)
Education	.004	.003
	(.007)	(.007)
Gender	.004	.005
	(.013)	(.012)
Age	-.012***	.003
	(.004)	(.065)
Rural	-.014***	-.015***
	(.004)	(.005)
Party ID	-.261***	-.225***
	(.0873)	(.089)
Reciprocity		-.223***
		(.065)
Constant	.204	.836
	(.497)	(.549)
N	1176	1107
Log-L	-636.67	-601.64

Standard errors are in parenthesis. Models 1 and 2 estimated using multi-level logit analysis. * denotes p<.10, ** P<.05, *** p<.01. Two-tailed test.

3.3 The Demand-Side of Vote Buying: Testing Hypotheses 2.2 and 2.3

3.3.1 Field Experiment

The hypotheses that emerge from the theoretical argument (Hypotheses 2.2 and 2.3) predict how information flows affect the individual-level vote behavior. Hence, an appropriate empirical research design requires the use of a carefully designed randomized field experiment. Indeed, carefully designed randomized experiments allow one to accurately test theories at the individual-level. (Green and Gerber 2003; Wantchekon 2003; Hyde

2007).[94] I thus test the hypotheses resulting from my theoretical argument using an original and carefully designed randomized field experiment conducted during the 2008 local elections in Nayarit, Mexico.

Field experiments have become an increasingly common method in American politics, but few scholars have used them in comparative politics (Wantchekon 2003; Guan & Green 2006; Humphreys et al 2006; Hyde 2007). Although both field and laboratory experiments test hypotheses by using random samples and assigning treatments in a controlled environment, field experiments can overcome one major criticism of laboratory designs—the problem of external validity, which states that the results of laboratory experiments are not necessarily generalizable to the 'real world'-. Field experiments take place in a natural setting, thus maximizing the chances that the results will indeed tell us something useful about the real world (Green & Gerber 2003).

Methodologically, the principal virtue of field experiments is that randomization provides a well-defined procedure for deriving unbiased assessments of causal inference. The estimations of field experiments are likely to be unbiased for two reasons. First, the independent variable in randomized field experiments is the only factor that varies systematically in the experiment. That is, if the experiment is appropriately controlled, the confounding variables are eliminated. Furthermore, field experiments allow researchers to develop a dependent variable that truly reflects the phenomenon under study (i.e. internal validity), and to minimize the possibility of measurement error (i.e. the variable can be measured with a high degree of reliability).

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[94] Other works involving the use of randomized field experiments are: Guan and Green (2006), Nickerson (2008), Humphreys et al (2006).

Two reasons explain why this field experiment focuses on Mexico. First, Mexico experienced a transition to a full-fledged democracy in 2000 (Magaloni 2006). Given its recent transition to democracy, both citizens and parties in Mexico are still learning the dynamics of voting, in particular, what constitutes a proper electoral practice. Thus the Mexican case provides an interesting laboratory to study how information flows affect voters' perceptions and behavior with respect to a widespread act of electoral corruption (i.e. vote buying) that may potentially harm democracy. Second, in the American politics context, several scholars have already documented that information flows are among the most important factors in forming attitudes and consequently affecting the political behavior of voters.[95] Hence, it is possible that the information supplied to citizens about the negative consequences of vote buying may influence the voters' decision to engage in vote buying, in a similar way that they influence other decisions about politics during their lives, such as political participation, voting behavior or the evaluation of elected officials. I describe below the field experiment used to empirically test hypotheses 2.2 and 2.3 of this dissertation.

3.3.2 Experimental Design

The field experiment employed in this study was conducted during the 2008 local elections in Nayarit, Mexico.[96] While Nayarit is only one of Mexico's 32 states, its socio-demographic, economic, and political indicators not only resemble those of the whole country, but also introduce an important degree of variation across and within different cases. Moreover, the timing of the field experiment allowed me take full advantage of the

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[95] See, for e.g., Converse (1964), Zaller (1992), Lavine et al (1996), Alvarez and Brehm (2002), Binzer-Hobolt (2005).

[96] Figure 1 details the geographical location of Nayarit, Mexico.

effervescent electoral climate lived during the 2008 Nayarit municipal and local legislative elections. Further, as Figure 3.2 illustrates, Nayarit has the second highest score of electoral volatility across Mexican states. Together, these factors allow me to generalize my findings to other parts of Mexico.[97]

The field experiment had three different stages. In a pre-electoral survey administered between May and June of 2008, we asked individuals about their interest in politics, party affiliations, and opinions on a number of economic and political issues. In this first stage of the experiment we randomly selected 9 electoral districts among the state's 18 using a multi-stage, stratified sampling design. To this end, we divided Nayarit into 5 regional strata: north, south, east (sierra), west, and central (Tepic).[98] Our goal was to allocate to each stratum sample shares that would be proportional to the total number of registered voters in the state.[99] This strategy yielded 17potential clusters (electoral districts) to choose from, among which 8 were randomly selected.[100] Figure 3.3 shows the geographical definitions of these strata. The next step was to randomly choose 8-10 electoral sections from each of the eight electoral districts. Within each electoral section, we randomly picked the households to

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97 More specifically, Nayarit has an estimated 2005 population of 920,185. The state in 2005 was 51 % female, 49 % male, 95% mixed, and 5% indigenous. In 2000, 60 % are aged 18 or older, and 6.2 % are 65 or older. The median per capita income in 2005 was $7,548. Politically, Nayarit has 20 municipalities and 18 electoral districts with more than 600,000 registered voters. As in the rest of Mexico, Nayarit's most important political parties are the PRI, PRD, and PAN. In the 2005 gubernatorial election, 46.2 % voted for the PRI, 42.3% for the PRD, 4.04% for the PAN, and 5.4% for other political parties. Turnout in the 2005 gubernatorial election surpassed 60%.

98 Tepic is the state capital. Its relative size with respect to other cities and its economic and political importance allowed me to consider treat it as an independent regional stratum

99 Author's calculations using data from the Instituto Electoral de Nayarit (IEEN). http://ieen.org.mx/

100 The electoral districts selected for this study are: District II and V (Tepic), District (Compostela), District (Jala-Ahuacatlan), District (Acaponeta-Huajicori), District (Tuxpan), District X (Santiago), District (La Yesca).

apply the survey.[101] The survey was carried out by the author and a team of research assistants.[102] The sample size of the pre-electoral survey is 513.[103]

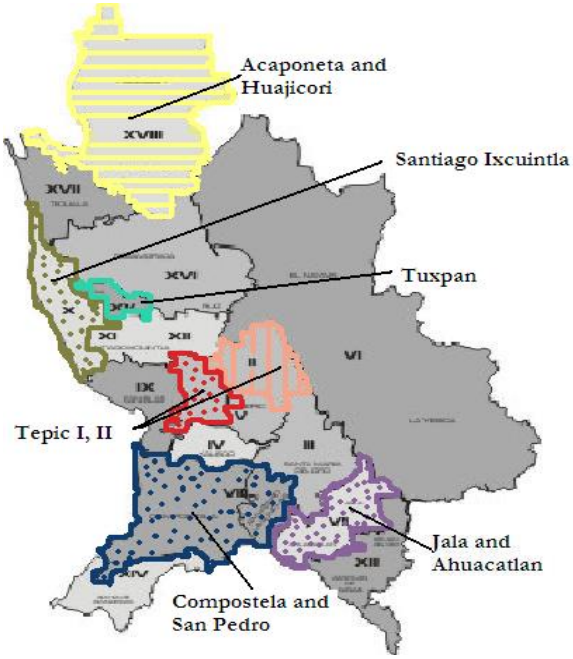

Figure 3.5: Geographical Definitions of the Strata

In addition to the economic, social and political issues covered in the pre-electoral survey, we asked respondents to recall if they were exposed to any type of political propaganda during the last three years. Individuals who responded negatively to this question were excluded from the study. Finally, my sample only includes individuals who

101 The sample selected in the pre-electoral survey is fairly representative in terms of size, geographical distribution, age, sex, and other economic and political indicators.

102 During the pre-electoral survey, the team of research assistants consisted mainly of undergraduates from the Autonomous University of Nayarit and University of El Bajio. All the research assistants completed a one week course on survey techniques.

103 The pre-electoral survey is available from the author upon request.

completed the pre-electoral survey; individuals who refused to answer to any of the questions in the survey were not assigned to a treatment or control group.

Individuals included were randomly assigned to a treatment or a control group. Individuals included in the sample are those who have been exposed to political propaganda, participated in the survey, and responded to all of the survey questions. The proportion of subjects placed in treatment and control groups was constant across strata. Overall, 175 individuals were included in the treatment group and 182 were assigned to the control group.[104] Table 1 contains the descriptive statistics for the complete experimental sample, the treatment, and control groups.

The experimental treatment in this design was three-week anti-vote buying campaign that informed individuals in the treatment group about the negative consequences of vote buying on democracy. To maximize the amount of information received by the individuals in the treatment group, two waves of treatment were administered. During the first wave, ten volunteer members of the Autonomous University of Nayarit distributed leaflets and offered a verbal explanation about their content. The leaflet, as mentioned before, informed individuals about the severity of vote buying in Nayarit and its negative consequences on democracy.[105] [106] While the volunteers did not have any previous experience canvassing voters, they were carefully trained and reported no problems in delivering the message. Conversations were very brief and all the individuals in the treatment group received a leaflet

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104 Due to accessibility constraints 42 other individuals who answered all the questions in the pre-electoral survey and who have been exposed to political propaganda were not included in the experimental sample.

105 The leaflet is presented in Figure 3.x

106 The distribution of leaflets was conducted during the first two weeks of June, 2008.

after the verbal explanation. The volunteers were also instructed to deliver leaflets to family members and neighbors of the treatment group.[107]

To maximize the amount of information successfully received by the individuals in the treatment group, we mailed another leaflet to the individuals in the treatment group two weeks before the election.[108] Appendix 1 contains the leaflets distributed and mailed before the elections, and the script used by volunteers during the first wave of treatment.

The final stage of this experiment included a post-electoral survey that was conducted two weeks after the elections. The response rate of the post-electoral survey was 91% for the treatment group and 89% for the control group. Besides providing additional information on socio-economic and political indicators, the post-electoral survey allowed me to obtain individual-level data on a wide range of topics including the citizens' perception of the frequency of vote buying, the likelihood to surrender money in exchange for votes, and information about the political parties that offered money in exchange for votes during the local election. In what follows, I explain the dependent, independent, and control variables, as well as the statistical methodology of this study.

TABLE 3.2

DESCRIPTIVE STATISTICS, FIELD EXPERIMENT

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107 While neighbors and family members received the leaflet, they did not receive a verbal explanation of the message.

108 Of course, a complication with the mailed material is whether the letters were delivered to the individuals in the treatment group. To account for this potential pitfall, we asked a randomly selected number of individuals from the treatment group whether they had received the mailed leaflet. More than 85% reported receiving the mailed material.

	Mean Complete Sample	Mean Treatment Group	Mean Control Group
Selling Votes	.195		
	(.397)		
Frequency of Vote Buying	2.13		
	(1.06)		
Income	3.59	3.76	3.41
	(1.99)	(.152)	(.161)
Housing	2.19	2.27	2.11
	(1.02)	(.083)	(.077)
Education	5.09	4.91	5.28
	(.706)	(.205)	(.158)
Gender	.431	.444	.418
	(.496)	(.391)	(.391)
Age	43.95	44.22	43.66
	(17.25)	(1.35)	(1.37)
Rural	.347	.271	.425
	(.477)	(.035)	(.039)
Party ID	2.21	2.13	2.28
	(.882)	(.079)	(.069)
	N=322	N=160	N=162

Standard Errors in parentheses

3.3.3 Variables and Statistical Analysis

3.3.3.1 Dependent Variable

Thisdissertation's theoretical section suggests that individuals exposed to information about the negative consequences of vote buying on democracy: 1) perceive vote buying as a more frequent phenomenon, and 2) are less likely to surrender their votes in exchange for money. Thus, to test the hypotheses derived from the theoretical argument, I make use of two dependent variables contained in the post-election survey: *Frequency of Vote Buying* and *Selling Votes*.

The dependent variable in hypothesis 1 is *Frequency of Vote Buying*. *Frequency of Vote Buying* is an ordinal variable ranging from 1 to 4. I operationalize this variable using the following question: "In your opinion, how frequently do you think political parties or candidates in your neighborhood offer money in exchange for votes on a scale from 1 (Never) to 4 (Always)?" The response of each is individual is thus coded using the following scale: 1) Never, 2) Rarely, 3) Sometimes, and 4) Always. The 1-4 ordinal dependent variable –*Frequency of Vote Buying*- extracted from the post-electoral survey measures the individual's perceptions about the frequency of vote buying in their neighborhood.

Selling Votes –the dependent variable of interest in hypothesis 2- measures the likelihood of respondents to engage in vote buying. To operationalize this variable, respondents were asked the following question: "If a political party offers you money in exchange for your vote, will you be willing to sell your vote?" Yes=1 or No=0. Responses to this question were coded dichotomously, where 1 indicates that the voter is willing to accept money in exchange for his or her vote and 0 indicates that not.

The dependent variables in this study differ in two main ways from those used in previous studies on vote buying and clientelism (Brusco et al 2004; Stokes 2005; Nichter 2008). First, rather than asking respondents if they received something from a candidate or party, my dependent variables provide information on whether individuals may be lured to vote for a party or candidate who offers money in exchange for votes, and their perceptions about the frequency of vote buying. Thus, the selection of my dependent variables –*Frequency of Vote Buying* and *Selling Votes*- allows me to make much more generalizable claims about the demand-side of vote buying. I see that it's different, but I don't understand why you consider this more generalizable or better in other respects.

Second, in the literature on vote buying and clientelism, vote buying entails an exchange of any type of material good (money, gifts, construction materials, jobs, etc.) between parties and/or candidates and citizens (Brusco et al 2004; Nichter 2008). Arguably, this is a valid definition of clientelism. Yet the exchange of money for votes (i.e. vote buying) is only one component of clientelism. In contrast with previous studies, I define vote buying as a purely monetary exchange between voters and politicians.[109] The uni-dimensional definition used in this study, in turn, allows me to: 1) more easily theorize about the causes and effects of vote buying, and 2) to solve any reliability problems with the estimators. OK, this is clearer.

3.3.3.2 Independent and Control Variables

I use one key independent variable to test the predictions of hypotheses 2.2 and 2.3: a dummy variable that accounts for the treatment effect described above –the campaign informing about the negative consequences of vote buying on democracy-. This variable is *Treatment*, which is coded as 1 for individuals in the treatment group and 0 otherwise. Following the theoretical argument, I hypothesize a positive relationship between the dependent variable *Frequency of Vote Buying* and the independent dummy variable *Treatment*. In addition, I expect a negative relationship between the dummy variable *Treatment* and my dependent variable *Selling Votes*. That is, I hypothesize that individuals in the treatment group will be less likely to accept money in exchange for votes.

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[109] More specifically, respondents were only asked whether they would accept 'money' in exchange for votes from a party or candidate. The DV *Frequency of Vote Buying* specifically asks: "How frequently do you think political parties and/or candidates offer 'money' in exchange for votes?"

Following the existing literature on vote buying and clientelism (Wantchekon 2003; Brusco et al 2004; Stokes 2005; Magaloni et al 2007; Nichter 2008), I control for several socio-demographic, economic, and political variables. Data for the control variables is drawn from both the pre-electoral and the post-electoral surveys. With respect to economic control variables, I include the ordinal variable *Personal Income*. Specifically, respondents were asked to select the range which best represents their household's monthly income. Answers to this question range from 1 (0 to 1299 MX Pesos) to 8 (8,000 or more MX Pesos). Following Stokes (2005) and Nichter (2008), I also include a variable that measures the quality of the houses inhabited by the individuals in this study: *Housing Quality*. The variable *Housing Quality* was assessed by the interviewer and ranged from 1 (poorest quality) to 4 (high quality).[110]

I also consider a set of socio-demographic control variables: *Age*, *Gender*, *Education*, and *Urban*. *Gender* is a dummy variable coded as 1 if the respondent is a male and zero otherwise. The individuals in the sample were also asked to select the category that best represents their educational level. The possible choices ranged from 1 (Illiterate) to 10 (Graduate degree). Hence, the variable *Education* is an ordinal variable ranging from 1 to 10. Data for the variables *Age*, *Gender*, and *Education* is from the post-electoral survey. Finally, I also control for the geographical location of the respondent. Specifically, the dummy variable *Rural* takes the value of 1 if the respondent lives in a rural area and zero if the individual lives an urban area. I code the dummy variable *Rural* using data from the National Institute of Statistics, Geography and Informatics (INEGI).[111]

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110 A more detailed description of the characteristics assessed by the interviewers is available upon request.

111 http://www.inegi.gob.mx

Finally, I make use of one political control variable: *Party ID*. Data for the ordinal variable Party Ideology is drawn from the post-electoral survey. To code *Party ID,* the participants in this experimental study were asked the following question: "Generally, would you consider yourself 1) Panista, 2) Priista, 3) Perredista, or Other?"[112] Thus, *Party ID* is an ordinal variable ranging from 1 to 4. Having described the empirical strategy, statistical methodology, and variables used to empirically test my theoretical claims, I now turn to present the findings and analysis of this study.

3.3.4 Findings and Analysis

The empirical strategy to test the hypotheses described in the theoretical section of this study is fairly straightforward. This is because the random or near-random assignment of individuals to treatment allows me to use fairly simple statistical techniques without compromising the accuracy of my results. I begin the statistical test of hypotheses 2.2 and 2.3 by estimating the difference in means between the treatment and control groups. More formally, the difference-in-means estimator can be expressed as follows:

$$E[y_k \mid TR=1] - E[y_k \mid C=1] \qquad (2)$$

where y_k is the value of the dependent variable, *TR* and *C* are dummy variables that take the value of 1 if the individual is exposed to the treatment and zero otherwise, and $E[Y_k]$ is the expectation operator for the dependent variables (that is, the average value of the variable across the whole population of interest).

112 Priista=Supporter of PRI; Panista=Supporter of PAN; Perredista=Supporter of PRD. PAN=Right, PRI=Center, PRD=Left.

The results from estimating equation (2) are reported in Table 3.4. Specifically, the difference in means estimation confirms the prediction of hypothesis 2.2 –that treated individuals (i.e. those exposed to information about the negative consequences of vote buying on democracy) will perceive vote buying as a frequent phenomenon. The results presented in Table 3.4 indicate that the mean of the dependent variable *Frequency of Vote Buying* is 2.02 and 2.25 for the treatment and control groups respectively.[113] In addition, the t-value of 1.96 suggests that the difference in means is statistically significant at the 95% confidence level. The aforementioned difference in means is depicted in Figure 3.4.

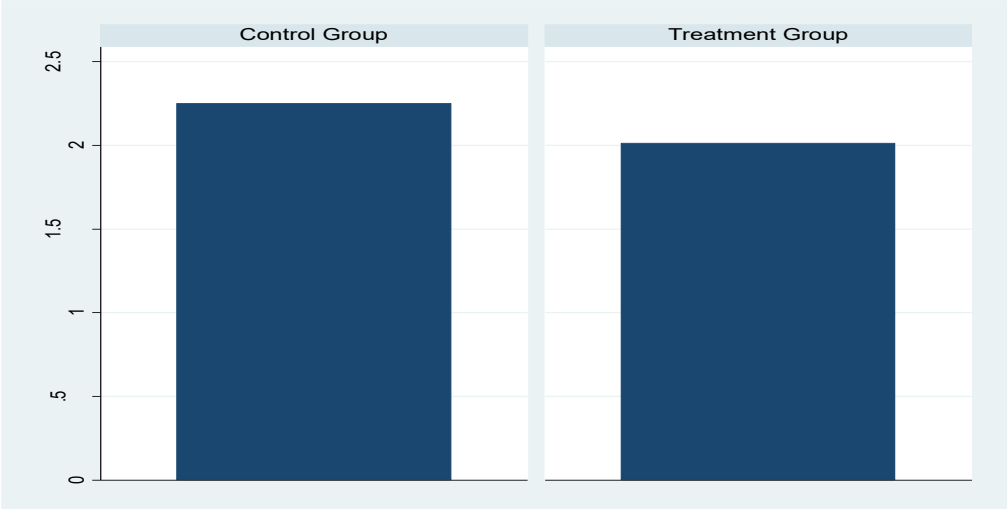
TABLE 3.3

MEANS-IN-MEANS ESTIMATION

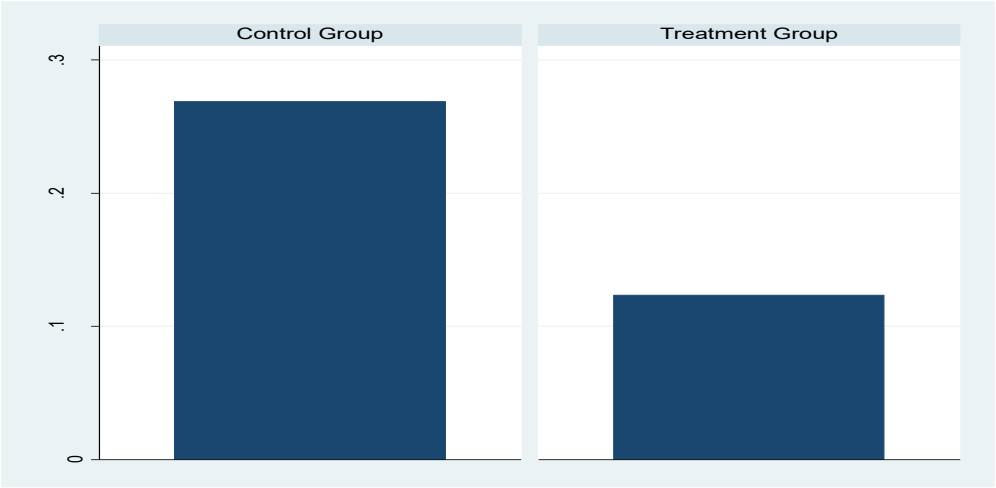
113 The standard errors are in parenthesis.

	DV: Frequency of Vote Buying	*DV: Selling Votes*
Treatment	2.012 (.081) N=152	.123 (.025) N=160
Control	2.25 (.089) N= 158	.268 (.035) N=162
Difference	.237 (.060) N= 310	.145 (.022) N=322
t statistic	1.96 **	3.33***

*** significant at 99%, ** significant at 95%, * significant at 90%. Sample sizes (N) are indicated below the means. Standard errors are in parenthesis.



A



B

Figure 3.6: Difference in Means for *Frequency of Vote Buying* (A). Difference in Means for *Selling Votes* (B).

Likewise, the mean of *Selling Votes* supports hypothesis 2.3. The means of the treatment and control groups are respectively .123 and .268. The results also indicate that the difference in means is statistically significant at the 99% confidence level (*t value*=3.33). Figure 3.4 displays the differences in the likelihood of *Selling Votes* between the treatment and control groups. Taken together, the difference in means and its statistical significance suggest that individuals in the treatment group are less likely to exchange votes for money. Hence, the empirical evidence presented above strongly supports the prediction in hypothesis 2.3.

Readers may suspect that the difference in means estimation is not comprehensive enough to make robust claims about the effects of the treatment effect on my dependent variables (*Frequency of Vote Buying* and *Selling Votes*). While I am confident about the accuracy of the difference in means estimations, I now present a more complete analysis of the treatment effect with multivariate analysis. The analysis provides an estimation of the treatment effects controlling for several socio-demographic, economic, and political

variables. Given the nature of my dependent variables, I estimate a series of multivariate Probit (DV is *Selling Votes*) and Ordered Probit (DV is *Frequency of Vote Buying*) analyses.[114]

Table 3.5 shows the effect of the information about the negative consequences of vote buying on democracy on the key dependent variables –*Frequency of Vote Buying* and *Selling Votes*-. For each set of regressions, I present 3 specifications. Models 3 and 6 include only the treatment group indicators. Models 4, 5, 7 and 8 include, in addition to the treatment effect, a series of control variables usually linked with vote buying.

To begin with, the coefficient on the main independent variable *Treatment* is positive and statistically significant at the 1% level in the baseline model 3. This result statistically corroborates hypothesis 2.2 of this dissertation. In other words, it suggests that the information about the negative consequences of vote buying on democracy sensitized voters about the frequency of vote buying, increasing the probability that they perceive vote buying as a pervasive and frequent phenomenon. Models 4 and 5, Table 3.6 present the results of the probit estimation when the socio-demographic, economic, and political control variables are included in equation (2).[115] The coefficient on *Treatment* remains positive and highly significant at the 1% level in both models, therein indicating that individuals who were exposed to the treatment effect are more likely to perceive vote buying as a frequent phenomenon.

Interestingly, the findings presented in models 4 and 5 suggest that only one control variable affects the voters' perceptions about the frequency of vote buying -*Income*-

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114 The results do not vary significantly when the estimations are performed using logit and ordered logit estimators.

115 Model 4 includes only economic and socio-demographic control variables. Model 5 includes all the economic, socio-demographic and political control variables.

neither predicts the voters' perception about the frequency of vote buying.[116] The negative and statistically significant (at the 1% confidence level) coefficient on *Income* suggests that, as the level of *Income* increases, voters perceive that vote buying is not frequently used as an electoral strategy by political parties. A possible explanation for this result is that parties and politicians often target poor individuals with monetary offers. This, in turn, makes them more likely to perceive that vote buying is a pervasive and frequent phenomenon in their neighborhood. The Wald tests derived from models 4 and 5 indicate that both *Treatment* and *Income* substantially improve the fit of the models (χ^2=8.56, 5.24 for these models, respectively).

TABLE 3.4

PROBIT AND LOGIT ESTIMATIONS, FIELD EXPERIMENT

116 The Wald test for model 4 indicates that including the economic and socio-demographic control variables in the probit model does not improve the fit of the baseline model 3 (χ2=2.53).

	Model 3 DV: Frequency of Vote Buying	*Model 4 DV: Frequency of Vote Buying*	*Model 5 DV: Frequency of Vote Buying*	*Model 6 DV: Selling Votes*	*Model 7 DV: Selling Votes*	*Model 8 DV: Selling Votes*
Treatment	.2402**	.260***	.268***	-.541***	-.508***	-.401***
	(.123)	(.128)	(.138)	(.165)	(.171)	(.183)
Income		-.074***	-.074***		.010	.017
		(.036)	(.038)		(.038)	(.041)
Housing		-.004	.022		-.034	-.009
		(.064)	(.067)		(.0861)	(.091)
Education		.0367	.029		.014	.0296
		(.029)	(.031)		(.0458)	(.048)
Gender		.034	.040		-.183	-.192*
		(.129)	(.137)		(.171)	(.124)
Age		-.003	.005		-.004	-.005
		(.003)	(.004)		(.005)	(.005)
Rural		-.069	-.107		.057	.123
		(.133)	(.142)		(.178)	(.189)
Party ID			-.074			.198***
			(.076)			(.099)
Constant				-.616***	-.298	-.859*
				(.106)	(.379)	(.482)
N	310	306	279	322	318	279
Log-L	-406.35	-397.37	-345.95	-153.67	-150.48	-129.71

*** significant at 99%, ** significant at 95%, * significant at 90%. Standard errors are in parenthesis. Models A, B and C estimated using ordered probit. Models D, E and F estimated via probit.

Turning now our attention to Models 6-8, table 3.6 where the dependent variable is *Selling Votes*, the coefficient on the main independent variable –*Treatment*- is negative and statistically significant at the 1% confidence level in the baseline model 6. The negative effect and statistical significance of Treatment does not vary in the additional specifications of the model (Models 7 and 8). This result confirms the prediction in hypothesis 2.3. Specifically, the results in Models 6, 7 and 8 suggest that citizens exposed to the information about the negative effects of vote buying on democracy are, on average, 40% less likely to engage in vote buying than those voters who did not receive the information about vote buying.

With the exception of *Gender*, in model 8, all other economic and socio-demographic control variables are statistically insignificant.[117] The coefficient on *Party ID* (Model 8, table 3.6) is positive and statistically significant at the 1% confidence level. This suggests that supporters of the left-wing parties are more likely to accept monetary offers from political parties or candidates. The result of the Wald test (χ^2=9.99) suggests that the coefficients on *Treatment*, *Gender*, and *Party ID* are statistically significant from zero, hence including these variables substantially improves the fit of the model.

3.3.5 Robustness Tests

Although I am fairly confident about the accuracy of the empirical results presented above, I now check their robustness by calculating a series of propensity score based estimators as suggested by Rosenbaum & Rubin(1983), Rosenbaum & Rubin(1984) and D'Agostino & Rubin(2000). The propensity score based estimators are useful to eliminate concerns of a non-random selection of the treatment and control groups (i.e. that the covariates have not identical distributions in the two groups).

More technically, the propensity score is the predicted probability of group membership from a logistic regression. It represents the propensity for an observation to be in either the treatment or the control group. By adjusting for the value of the propensity score, one effectively adjusts for any group differences attributable to the variables used to create the propensity score. Moreover, the values of the propensity scores can serve as a diagnostic to evaluate the comparability of the groups in a quantitative way (Rosenbaum & Rubin 1983).

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117 The negative and statistically significant (at the 10% confidence level) coefficient on Gender suggests that, holding all other variables constant, men are more likely to exchange votes for money.

TABLE 3.5

PROPENSITY SCORE BASED ESTIMATION, FIRST STAGE

	Model 9 *DV: Treatment*	*Model 10* *DV: Treatment*
Income	-.064*	-.039
	(.040)	(.043)
Housing	.151***	.144
	(.074)	(.078)
Education	-.021	-.014
	(.033)	.036
Gender	.081	-.010
	(.147)	(.157)
Age	.000	-.001
	(.004)	(.004)
Rural	-.473***	-.592***
	(.159)	(.161)
Party ID		-.104
		(.088)
Constant	.119	-.399
	(.321)	(.402)
N	318	279
Log-L	-211.96	-182.94

*** significant at 99%, ** significant at 95%, * significant at 90%. Standard errors are in parenthesis. Models 7 and 8 are Probit estimations.

The results of the propensity scores are presented in models 11-14, Table 3.7. The positive and highly significant coefficient on *Treatment* in models 11-14 confirms the robustness of the results of models 3-5, Table 3.7. The results in models 11 and 12, table 3.6 suggest that individuals in the treatment group perceive vote buying as a more frequent phenomenon in their neighborhood than those in the control group (hypothesis 2.2). Moreover, the statistical results in models 13 and 14 corroborate the predictions of hypothesis 2.3 –that individuals exposed to the treatment are less likely to exchange money

for votes-. Models 11-14 confirm that the results presented in the previous section are not driven by inaccuracies in the randomization process, confirming their statistical robustness.

TABLE 3.6

PROPENSITY SCORE BASED ESTIMATION, SECOND STAGE

	Model 11 DV: Frequency of Vote Buying	*Model 12 DV: Frequency of Vote Buying*	*Model 13 DV: Selling Votes*	*Model 14 DV: Selling Votes*
Treatment	-.254***	-.261***	-.503***	-.394***
	(.127)	(.137)	(.169)	(.181)
Propensity	-.447	-.622	-.431	-.747
	(.560)	(.524)	(.744)	(.701)
Constant			-.419	-.293
			(.369)	(.364)
N	306	268	318	279
Log-L	-399.64	-348.88	-151.61	-131.83

*** significant at 99%, ** significant at 95%, * significant at 90%. Standard errors are in parenthesis. Models 9 and 9 are Probit estimations. Models 11 and 12 estimated using Ordered Probit.

3.4 Conclusion

Recent studies on vote buying have examined how poverty (Kitschelt 2000; Cornelius 2004; Stokes 2005; Magaloni 2006), reciprocity (Brusco et al 2004; Finan and Schechter 2009), or the level of intra-party competition (Hicken 2002, 2007) affect vote buying. In contrast with those studies, the theory presented in the previous chapter of this dissertation suggests that the level of institutionalization of the party system affects both the candidates' and voters' vote buying behavior. Using the results from an original survey on vote buying and clientelism in Mexico (H 2.1) and an original randomized field experiment

(H 2.2 and H 2.3), this chapter has presented a cumulative body of evidence supporting hypotheses 2.1, 2.2, and 2,.3.

First, the estimates from multi-level logit models suggest that electoral –that operationalizes the level of party system institutionalization- has a positive and statistically significant effect on vote buying. Therefore, they provide evidence support for hypothesis 2.1. Second, results from the field experiment indicate that voters who are exposed to information flows about the consequences of vote buying are less likely to engage in this corrupt electoral practice. As such, the findings presented in this paper have three important implications. On the one hand, the quantitative evidence supporting hypothesis 2.1 suggests that certain characteristics of domestic political institutions can affect the likelihood of vote buying. Thus scholars need to devote more attention to analyzing what other institutional factors prevent or foster this phenomenon.

On another, policymakers can also benefit from the results presented in this chapter. The results presented here suggest that political information about the negative consequences of vote buying generates a hostile environment to this phenomenon. Hence, governments in the developing world interested in curtailing vote buying would benefit from increasing the degree of media freedom in their countries. More media freedom may provide citizens with accurate information about the frequency and severity of vote buying and its potential consequences for their individual and collective democratic rights, helping to fight corrupt electoral practices that endanger democracy in the developing world.

Notwithstanding the implications of this study, more work needs to be done to fully comprehend the determinants of vote buying. More specifically, the research presented in this paper can be extended along the following lines. First, it would be worthwhile to investigate how and to what degree information about the negative consequences of vote

buying affects the electoral gains of political parties that practice it. If citizens learn and internalize norms against vote buying, they may punish those who offer money in exchange for votes at the polling booths, reducing the electoral returns of vote buying. Second, any inferences from this study about the effects of information flows on vote buying behavior should recognize that the results may depend on several specific features of this field experiment such as the choice of subjects, length and timing of the study, and the choice of messages. To address any challenges involving the external validity and generalization of the results presented in this paper, it would thus be interesting to design and implement additional field experiments in other developing countries.

CHAPTER 4:

PARTY SYSTEMS AND ELECTORAL FRAUD IN LATIN AMERICA, 1982-2006

4.1 Introduction

I suggested in chapter 2 that the level of party system institutionalization plays a critical role in explaining the variation of vote buying across Latin American countries. More specifically, I argued that weakly institutionalized party systems generate patterns of political competition that increase the uncertainty of politicians about the outcome of electoral processes. This, in turn, lures candidates and political parties to use vote buying rather than political platforms to secure electoral victories. I also suggested more generally in chapter two that the effects of the degree of party system institutionalization on the frequency of vote buying has important consequences for aggregate political outcomes. More specifically, hypothesis 2.4 in chapter two posits that by increasing the probability of vote buying, weakly institutionalized party systems increase the likelihood of aggregate electoral fraud in those countries.

In this chapter, I statistically test this hypothesis on a comprehensive time-series-cross-sectional dataset of 18 Latin American countries between 1982 and 2006.[118] The results of a series of logit models indicate that the level of institutionalization of the party system –

[118] Specifically, these countries are: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, the Dominican Republic, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay, and Venezuela. Cuba is excluded from the sample due to a lack of democratic elections during this period.

operationalized as the level of electoral volatility for each legislative election in each country- has important effects on the likelihood of electoral fraud across Latin American countries thus providing strong empirical support for hypothesis 2.4. More specifically, higher levels of electoral volatility (which indicate low levels of party system institutionalization) increase the likelihood of electoral fraud in the countries included in my sample.[119] [120]My results remain highly robust not only when I use different specifications of the model, but also when I employ different econometric estimators.

The empirical results presented in this chapter have several important implications for the literatures on electoral corruption and party systems in developing democracies. Most previous studies on electoral fraud suggest that economic and sociological factors are the primary determinants of this phenomenon (Auyero 1999; Trejo and Aguilar-Rivera 2002; Alston and Gallo 2009; Eisendstadt 1999; Ziblatt 2009). My results indicate, however, that the likelihood of electoral fraud in Latin America can best be explained by political factors (i.e. the level of instititutionalization of the party system). Hence, my findings suggest that future analyses of this phenomenon should explore in more detail the linkages between this act of corruption and other political institutions. In addition, the results presented in this chapter confirm that by increasing the likelihood of electoral fraud, weakly institutionalized party systems can indeed have devastating effects for democratic politics.[121]

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[119] In substantive terms, the likelihood of electoral fraud increases from 1% to 33% when electoral volatility goes from its minimum to its maximum value.

[120] The dependent variable, electoral fraud, measures whether vote fraud or candidate intimidation were serious enough to affect the outcome of the elections

[121] This intuition has been suggested before by scholars such as Mainwaring and Scully 1995; Moser 1999, 2001; Stoner Weiss 2001; and Mainwaring and Torcal 2005. However, this chapter is the first systematic test of this argument.

The rest of the chapter is organized as follows. In the next section, I provide a detailed description of the sample, the statistical methodology, the operationalization of the dependent variable, independent as well as control variables that are employed for testing hypothesis 2.3. I then present the results from testing this hypothesis. The chapter ends with a conclusion in which I summarize my results and the main contributions of this analysis to the existing literature.

4.2 Testing Hypothesis 2.4

4.2.1 Sample

In the theoretical chapter of this dissertation, I hypothesized that low levels of party system institutionalization increase the likelihood of electoral fraud in Latin America (hypothesis 2.4). To test the aforementioned prediction, I compile an original time-series-cross-sectional (TSCS) dataset of 18 Latin American countries between 1982 and 2006. Since my theory is not limited to this time period, I do not expect that extending the data to previous years would drastically change my inferences.[122] Therefore, my spatial unit of observation is the country. To leverage available country-level data, I use the year as my temporal unit of observation, resulting in a country-year unit of observation. My spatial domain is comprehensive as it includes all the Latin American countries for which data is available to carefully operationalize both the dependent, independent and control variables of interest to test hypothesis 2.4.

[122] However, extending the temporal domain to the years previous to 1982 is a very difficult task since election records are not easily available for all the democratic elections that occurred in Latin America before 1982.

The lack of competitive elections in the Latin American region before 1982 prevented me from extending the temporal domain of my sample.[123] However, the fact that my sample is comprehensive in both the temporal and spatial domains between 1982 and 2006 is advantageous because unlike previous studies of electoral fraud that mainly focus on single-case studies (Alston and Gallo 2009; Lehoucq 2002; Trejo and Aguilar-Rivera 2002; Eisendstadt 1999; Dominguez and McCann 1996), I am able to make much more generalizable claims if my hypothesis is supported by the empirical evidence. A second advantage of my sample is that it contains substantial cross-sectional and temporal variation in the dependent and independent variables, therein allowing me to test hypothesis 2.3 more effectively.

4.2.2 Dependent Variable

The aim of this chapter is to link the degree of party system institutionalization with the incidence of electoral fraud. Electoral fraud comes in at least three varieties: *coercion* and threats from state officials or political parties to induce voters to vote for a particular candidate or party (Kousser 1974; Posada-Carbo 1996; Anderson 2000); *vote buying* to increase or decrease the number of votes and voter turnout (Schaffer 2008; Nichter 2008); or systematic procedural violations, including, vote rigging, closing of poll stations early, manipulation of voter-registration rolls, and the failure to advertise elections or to distribute ballots in certain constituencies. The multidimensional nature of electoral fraud represents an important measurement challenge, particularly when one tries to operationalize this variable across space and time.

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[123] Cuba is excluded from the analysis because competitive elections did not take place in this country.

I circumvent this problem using a broader definition of electoral fraud. In the theoretical chapter of this dissertation, I followed Lehoucq (2003) and conceptualize *electoral fraud* as any electoral "violation of the law" that may or may not be practiced by incumbents in Latin American countries with some probability.[124] Lehoucq's definition is very useful for the purposes of this chapter as it allows me to measure my dependent variable, namely the phenomenon of *electoral fraud*, across Latin American countries between 1982 and 2006 using the World Bank's (2008) *Domestic Political Institutions* (DPI) variable electoral fraud.[125]

The World Bank's DPI measure of *electoral fraud* measures whether vote fraud or candidate intimidation were serious enough to affect the outcome of the elections. It captures extra-constitutional irregularities in electoral processes. Specifically, for each Latin American country in this study, the dependent variable –labeled as *electoral fraud*- is coded as "0" for countries where, for example, opposition parties are officially or constitutionally banned or where irregularities are not reported (although may still exist). Electoral fraud takes the value of "1" when opposition is officially legal but suppressed anyway.[126] It also takes the value of "1" when the electoral results are rigged by the incumbent or when opposition parties boycott the election because of intimidation of the government. The DPI codes this measure based (partly) on reports of electoral malfeasance or fraud provided by the US State department, Human Rights Watch, OSCE (Europe), the UN and the

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[124] Schedler (2002: 105) provides another broader definition of electoral fraud. Specifically, he defines electoral fraud as the "introduction of bias into the administration of elections such that the voting process itself is distorted."

[125] This measure of electoral fraud has been previously used in studies conducted by Hyde and Bailleau (2009) and Simpser (2004).

[126] Recording is irrespective of whether only opposition claims of fraudulent elections have occurred or whether allegations are backed by independent international observers.

Organization of American States. Irregularities in elections are coded in the DPI database regardless of whether or not opposition parties claim fraudulent elections.

The bar graph in figure 3.1 shows the frequency of electoral fraud that has occurred across Latin American countries in my sample (1982-2006). This figure shows that the frequency of electoral fraud in Latin American countries is not only reasonably common but also varies significantly. For example, in Nicaragua and Panama, the mean of electoral fraud is .63 and .6 respectively. In Peru, Guatemala, Panama, and Paraguay the mean of the dependent variable –*electoral fraud*- is substantially higher than in countries such as Argentina, Colombia and Mexico. The mean of electoral fraud across Latin American countries is .123 and its standard deviation is .329.

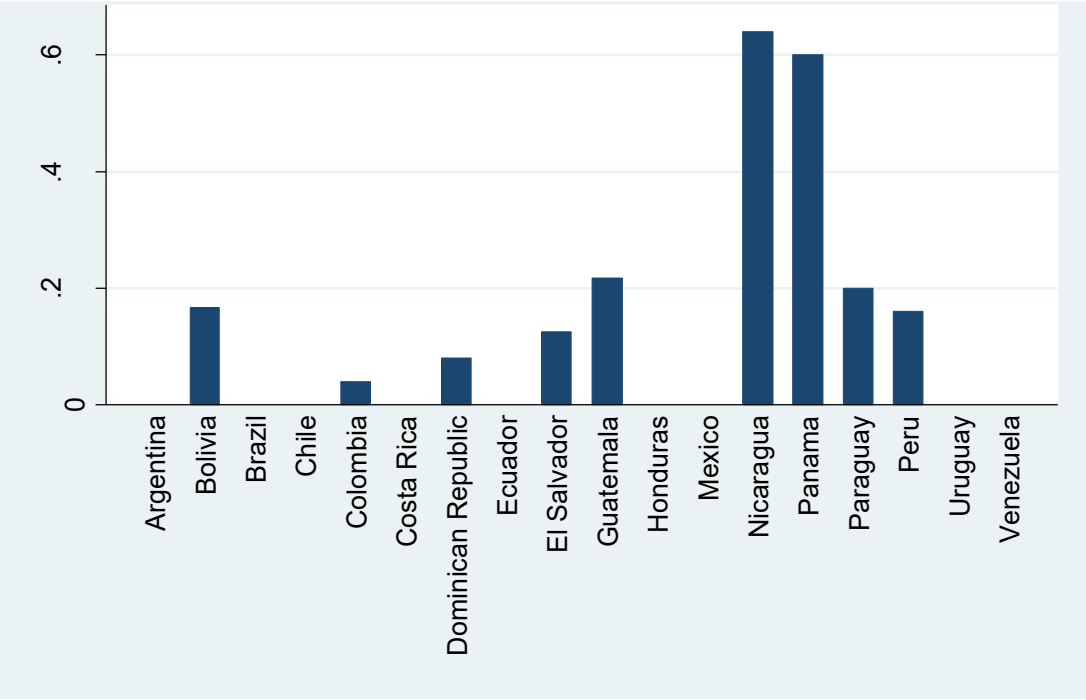

Figure 4.1: Electoral Fraud in Latin America, 1982-2006

4.2.3 Statistical Methodology

Given the structure of my dataset, I employ a time-series-cross-section (TSCS) logit model with random effects with controls for time dependence and autocorrelation, as suggested by Beck et al (1998).[127] Hypothesis 2.3, predicts a positive relationship between electoral volatility and the probability of electoral fraud. Given the prediction in the aforementioned hypothesis, it is reasonably safe to assume that the probability of fraud varies monotonically and remains within the bounds of [0,1].[128] In the logit model the error term is assumed to have a *standardized logistic distribution*. This distribution has a mean of 0 and a variance of 1 and is very similar to a normal distribution with the same mean and variance (Maddala 1983).

The introduction of controls for time dependence follows Beck et al (1998). According to the aforementioned authors, controlling for time dependence is important when estimating ordinary logit models because the estimates may be biased if the baseline hazard rate is time dependent. This potential problem, however, is avoided by introducing a series of temporal dummies in the models. Another possibility is to include cubic splines as controls for time dependence. But, given the data used in this study, both solutions –i.e. temporal dummies and cubic splines- present some econometric challenges. The inclusion of temporal dummies, for instance, not only uses many degrees of freedom but may also perfectly predict an outcome and drop those observations from the analysis. Likewise,

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[127] I also estimated all the models presented in tables 3 and 4 using probit models. The statistical significance and direction of the independent and control variables remain unchanged, suggesting that my results are robust even when I use a different estimator.

[128] See Ladd (1966).

including cubic splines in my models may lead to a perfect prediction of the dependent variable.

To circumvent these potential problems, I follow Wright (2008) and simply include time and various exponential transformations of the time variable in the models. The logit models presented below include the variables time, time-squared, and time-cube (Wright 2008: 232).[129] Moreover, I control for the potential autocorrelation effects by including a lagged dependent variable as an explanatory variable in the all the models presented in tables 3 and 4.

4.2.4 Independent Variable

Hypothesis 2.3 predicts a negative relationship between the level of party system institutionalization and my dependent variable across Latin American countries –i.e. higher levels of party system institutionalization decrease the likelihood of electoral fraud-. I thus need one key independent variable to test this prediction: the level of party system institutionalization for each country between 1982 and 2006. I now turn to describe the procedure to operationalize this variable for each country in my sample.

As in the previous chapter, *electoral volatility* measures the system-level change in party vote percentage from one election to another (Roberts and Wibbels 1999: 580). Therefore, the main independent variable in this chapter –*electoral volatility*- is measured using formula (1) in chapter 3. :[130] [131] [132] A detailed list of the legislative elections that were used to calculate

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[129] Likelihood ratio tests suggest that these time transformations are jointly significant. Thus, they need to be included as controls for time dependence.

[130] The first election after the onset of democracy in each country serves as the baseline for calculation.

the index of electoral volatility is presented in table 4.1. To this end, I relied on multiple primary and secondary sources to obtain the electoral results of the countries in my data. These sources include: (i) The Political Database of the Americas,[133] (ii) International Democracy and Electoral Assistance,[134] (iii) Latin American Network Information Center,[135] (iv) Nohlen (1993), (v) Instituto Federal Electoral (Mexico), (vi) The *Europa World Year Book*, and the (vii) *International Almanac of Electoral History*.[136] Using data from the aforementioned sources, the electoral volatility across Latin American countries has a mean of 26.27 and a standard deviation of 17.92.[137]

TABLE 4.1

DEMOCRATIC ELECTIONS IN LATIN AMERICA, 1982-2006

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[131] Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, the Dominican Republic, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay, and Venezuela.

[132] The formula is: $$Volatility(V_t) = \frac{\sum_{i=1}^{n} |p_{i,t}|}{2}$$

$$0 \leq V_t \leq 100$$

[133] Available at: http://pdba.georgetown.edu/Elecdata/elecdata.html.

[134] See: http://www.idea.int

[135] For this see: http://www1.lanic.utexas.edu/info/newsroom/elections/

[136] Mackie and Rose (1989).

[137] The minimum value is 3.5 (country, year here) and the maximum is 100 (country here).

Country	Elections in sample	Country	Elections in sample
Argentina	1983, 1985, 1987, 1991, 1993, 1995, 1997, 1999, 2001, 2003, 2005	Guatemala	1985, 1989, 1993, 1997, 2001, 2005
Bolivia	1985, 1989, 1993, 1997, 2001, 2005	Honduras	1985, 1989, 1993, 1997, 2001, 2005
Brazil	1986, 1990, 1994, 1998, 2002, 2006	Mexico	1982, 1985, 1988, 1991, 1994, 1997, 2000, 2003, 2006
Chile	1993, 1997, 2001, 2005	Nicaragua	1990, 1996, 2002
Colombia	1982, 1986, 1990, 1994, 1998, 2002, 2006	Panama	1989, 1994, 2000, 2005
Costa Rica	1982, 1986, 1988, 1990, 1992, 1994, 1996, 1998, 2000, 2002, 2004	Paraguay	1983, 1988, 1993, 1998, 2003
Dominican Republic	1982, 1986, 1990, 1994, 1998, 2002, 2006	Peru	1985, 1990, 1995, 2000, 2005
Ecuador	1984, 1986, 1988, 1990, 1992, 1994, 1996, 1998, 2000, 2002, 2004, 2006	Uruguay	1989, 1994, 1999, 2004
El Salvador	1985, 1990, 1995, 2000, 2005	Venezuela	1983, 1988, 1993, 1998, 2003

The results presented in figure 5.2 suggest that the main independent variable –i.e. *electoral volatility*- varies substantially across Latin American countries. For example, in countries such as Bolivia (37.05), Brazil (35.50), the Dominican Republic (33.95), Ecuador (33.92), Guatemala (45), Peru (47.78) and Venezuela (31.95), the mean of electoral volatility is significantly higher than that in my whole sample. In contrast, electoral volatility in Chile (16.13), Honduras (6.85), Mexico (15.31), Paraguay (16.04), and Uruguay (14.35) is lower than 26.27, which is the mean across Latin America. These results are in line with previous

studies analyzing the causes of electoral volatility in this continent (Roberts and Wibbels 1999; Madrid 2005; Mainwaring 1998).

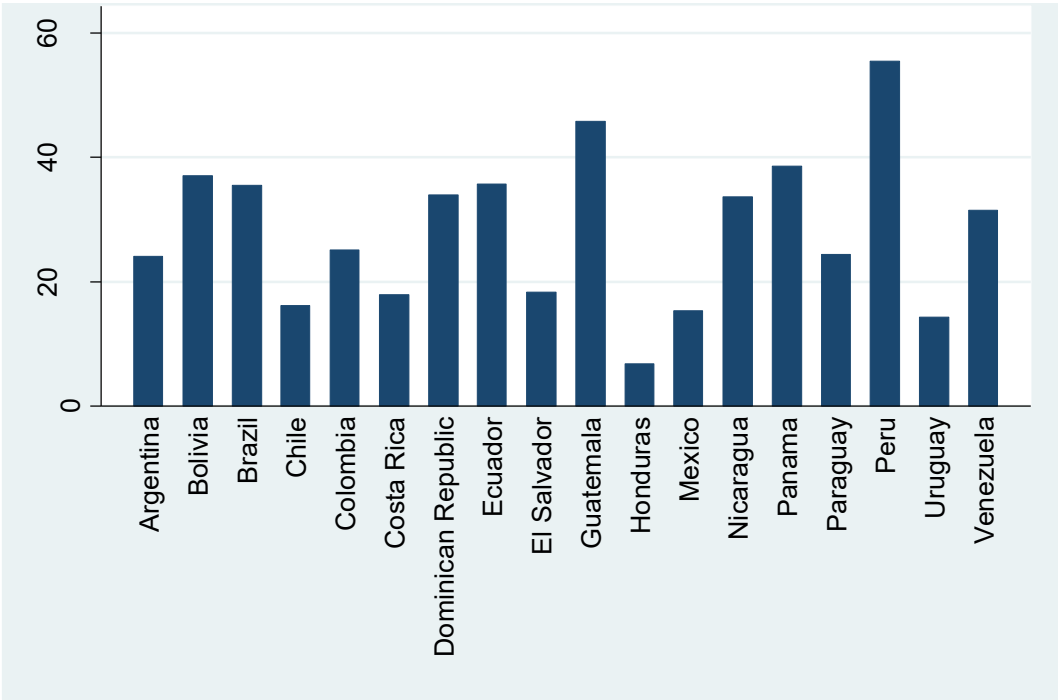

Figure 4.2: Electoral Volatility in Latin America, 1982-2006.

4.2.5 Control Variables

I include several control variables that have been identified by scholars as important determinants of electoral fraud in both developed (Ziblatt 2009; Anderson 2000) and developing countries (Dominguez and McCann 1996; Eisenstadt 1999; Varela Ortega 2001; Schaffer 2002; Schedler 2002; Lehoucq 2003; Simpser 2005; Hyde and Bailleau 2009; Milner et al 2010). Introducing these control variables in the empirical model is important as it allows me to test several alternative explanations presented by researchers who have already studied the causes of electoral fraud. To begin with, several scholars have already documented that higher levels of economic development makes it harder to political parties

to buy votes which, in turn, discourages electoral fraud (Auyero 1999; Shaffer 2002; Schedler 2002). I thus include the *log of GDP per capita* in the models estimated in this chapter. Following the aforementioned scholars, I hypothesize that higher levels of economic development decrease the likelihood of electoral fraud across Latin American countries. Data for *log of GDP per capita* is drawn from the Penn World Tables 6.1.

Dominguez and McCann (1996) and Ziblatt (2009) suggest that electoral fraud is more common in urban than in rural areas. This is because more urbanized areas encourage the formation of social networks that tend to give rise to more sustainable acts of mobilization and a greater willingness to protest a wide range of issues, including, perhaps, the outcomes of elections (Ziblatt 2009: 9).[138] I thus include the control variable *Urbanization*, operationalized as the percentage of a country's population living in urban areas.[139] Data for this variable is drawn from the World Bank's World Development Indicators (2003, 2008).

Scholars have also suggested that political factors explain the incidence of fraud. For instance, the partisan makeup of the incumbent government might determine how the election petitions are evaluated. The electoral manipulation of incumbent governments has indeed been extensively documented in several Latin American countries. For example, in Argentina, Botana (1979) argues that incumbent manipulation of the electoral system preserved the conservative's control of a political system until 1916, when the opposition Radical Civic Union won the general election. Moreover, Eisenstadt (1998) finds that infractions of electoral law picked up in Mexico when the PRI's domination began to crumble. To test the effects of partisanship on electoral fraud, I include the variable

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[138] In Mexico, Calderon and Cazes (1996) and Cazes (1996) find that violations of the secret ballot are more common in rural than in urban areas.

[139] Urban areas are cities with more than 1 million people (WDI, 2008).

partisanship which is drawn from the World Bank's (2008) Database of Political Institutions (1998). *Partisanship* is coded from a left to right scale (1-3 scale), where 1 denotes an incumbent government with a rightist ideology and 3 corresponds to a left-wing incumbent government.

Numerous scholars have argued that fragile political institutions enhance the possibility of electoral fraud as it makes it easier for incumbents to manipulate elections (Alston and Gallo 2009; Schedler 2006). I thus introduce three variables in my models to test this possibility. First, I include the variable democracy. I operationalize *polity* using the continuous measure from the Polity IV dataset (Gurr, 1989; Ward and Gleditsch 2006; Marshall and Jaggers 2002). This measure of democracy is not only commonly used in the literature that studies political freedom but also is well suited to capture gradual strengthening or weakening of democracy.[140] The Polity IV scores democracy on a scale of -10 (for the most autocratic regime) to 10 (for the most democratic regime) by three main criteria: executive constraints, openness of executive recruitment, and the competitiveness of political participation (Marshall and Jaggers 2002). Rather than measure separately the effects of autocracy and democracy, I adopt the growing convention of combining the two scales into a single index, DEMOC-AUTOC, which runs from -10 to 10, with 10 being the most democratic (Ward and Gleditsch 2006; Elkins 2000; Baum and Lake 2001; Li and Reuveny 2003).

I include the variable *new democracy*, which is a dummy variable taking the value of 1 in the year (*t*) in which a democratic transition occurred in each country *and* the *subsequent*

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[140] A distinction that is fundamental in the developing world and not captured in important works of democracy such as Przeworski et al (2000).

four post-transition years (t + 4) for each country in my sample that experienced a democratic transition; it is coded as zero for all other years. Put differently, *new democracy* is coded as 1 in the year of democratic transition and for the immediate four post-transition years – this leads to a total of five years – for each country in the data that experienced a democratic transition (Milner et al 2010). To test the effects of fragile democratic institutions on electoral fraud, I also include a variable that measures the degree of professional independence of the electoral management bodies in each country, labeled as *EMB*.[141] To code this variable, I rely on Hartlyn, McCoy and Mustillo (2008) who use the number of political actors involved in appointing its staff, the ratio of tenure of EMB staff to tenure of its appointers, and the relevance of the judiciary or civil society in nominating EMB officials to create an index of *EMB*. This index ranges from 1 to 10, and I code it here so that higher values correspond to higher independence.[142]

To assess the mobilization hypothesis which states that the degree of social mobilization might affect the inclination to file an organized protest against illegal elections (Ziblatt 2009), I include the proxy variable *turnout*, which is operationalized as the election turnout rate as a percentage of eligible voters for each constituency and each election in my sample. Data for turnout is drawn from the Latin American Network Information Center and the Georgetown's *Political Database of the Americas*. Following Ziblatt (2009), I expect a positive relationship between *turnout* and my dependent variable –*electoral fraud*-.

Other studies of *electoral fraud* suggest that this phenomenon is largely a function of sociological factors (Trejo and Aguilar-Rivera 2002; Lehoucq and Molina 2002). For

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[141] Hartlyn et al (2008) also provide an index of formal authority. My results remain robust even when I use the index of formal rather than profession autonomy in the regressions.

[142] The index of EMB is available for all the countries included in this analysis.

example, in a well-crafted study of electoral manipulation in Mexico, Trejo and Aguilar-Rivera (2002) argue that voting irregularities during the 1997 federal elections in Mexico are positively related to the monolingual share of the population, the effective number of ethnic groups, gender discrimination and the effective number of religions.[143] To control for the potential impact of sociological factors on election fraud, I include two control variables: *ethnic fractionalization* and *religious fractionalization*. *Ethnic Fractionalization* measures the likelihood that two people chosen at random will be from different ethnic groups (Easterly and Levine 1997).[144] Likewise, the variable *religious fractionalization* measures the probability that two people chosen at random will be from different religious groups.[145] Data for these variables is drawn from Przeworski et al (2000). A detailed description of the descriptive statistics of the dependent, independent and control variables is presented in Table 4.2

TABLE 4.2

DESCRIPTIVE STATISTICS

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[143] Trejo and Aguilar-Rivera (2002) also find that elections are negatively related to geographic dispersion.

[144] The ethnic fractionalization variable is calculated using a simple Herfindhal concentration index.

[145] The results remain robust when I include the percentage of catholic population rather than religious fractionalization in the models.

Variable	Mean	Standard Dev.	Minimum	Maximum
Electoral Fraud	.123	.329	0	1
Electoral Volatility	28.54	19.05	3.5	100
Polity	5.48	5.15	-8	10
New Democracy	.103	.304	0	1
Partisanship	1.52	.693	1	3
Turnout	67.34	22.23	21.3	94.7
EMB	3.36	2.39	1	8
Margin	13.99	11.47	.32	58
Log GDP	8.59	.412	7.67	9.44
Urban	61.93	16.58	33	91
BOP Crisis	.298	.457	0	1
Eco Crisis	.041	.199	0	1
ELF	.428	.183	.132	.743
Religious Frac	.173	.167	.019	.598
Inequality	50.84	5.53	36.85	62.1
Outsider	.077	.268	0	1

4.3 Results

Models 1-4 in table 5.1 report the results for testing hypothesis 2.3 for the dependent variable *electoral fraud*. As mentioned before, these models are each estimated using panel logit estimators with a lagged dependent variable, controls for time dependence and random effects. Specifically, Model 1 in table 5.1 reports the results obtained from a specification where in addition to the main independent variable *electoral volatility*, I included four variables that control for time dependence and potential autocorrelation: *time*, *time2*, *time3*, and *lag fraud*. The effect of *electoral volatility* in model 1 is positive and statistically significant at the 1% level in the panel logit specification which is estimated with random effects. Model 2, table 5.4, reports the results of a basic specification where in addition to the aforementioned variables I include the following economic, political and sociological control variables: *polity*, *log gdp*, *urban*, *elf*, and *religious fractionalization*. The estimated effect of electoral volatility on my

dependent variable electoral fraud is once again positive and statistically significant at the 1% confidence level. I add *new democracy* and *partisanship* to the specification in model 3, while in model 4 I include both the additional control variables mentioned above as well as *turnout* and *EMB*. In models 3 and 4, the effect of electoral volatility on the dependent variable remains positive and highly significant (1% confidence level).

TABLE 4.3

PANEL-LOGIT ESTIMATIONS: ELECTORAL VOLATILITY AND ELECTORAL FRAUD IN LATIN AMERICA 1982-2006

	1	2	3	4
Lag Fraud	4.88*** (.585)	4.53*** (.625)	4.61*** (.778)	5.05*** (1.11)
Electoral Volatility	.043*** (.014)	.048*** (.018)	.060*** (.019)	.092*** (.029)
Polity		.009 (.121)	.114 (.136)	.052 (.179)
New Democracy			.494 (1.40)	.033 (1.50)
Partisanship			-1.88*** (.805)	-2.58*** (1.13)
Turnout				10.7* (6.32)
EMB				-.373 (.346)
Log GDP		-1.15 (1.11)	-1.43 (1.26)	-1.72 (1.93)
Urban		-.009 (.035)	-.003 (.035)	-.008 (.053)
ELF		-2.17 (1.97)	-2.04 (2.23)	-2.00 (2.92)
Religious Fract.		1.08 (1.99)	.500 (2.00)	3.47 (3.67)
Time	.737 (.696)	.327 (.846)	.192 (1.01)	-.338 (1.15)
$Time^2$	-.052 (.052)	-.013 (.071)	.013 (.093)	.056 (.113)
$Time^3$	.001 (.001)	.000 (.001)	-.001 (.002)	-.002 (.003)
Constant	-8.02*** (.3.05)	4.07 (9.60)	7.17 (10.7)	4.86 (12.8)
Wald (x2)	72.04	57.65	43.67	33.07
Log Likelihood	-52.02	-44.51	-37.19	-27.76
N	382	323	316	308

***, ** and * denotes significance at the 1%, 5% and 10% level, respectively. The panel-logit models in this table are each estimated with random-effects.

The positive and statistical significant coefficients on electoral volatility in models 1-4, table 4.3, provide strong support for hypothesis 2.3. Yet they tell very little about the

substantive effect of *electoral volatility* on the main dependent variable –*electoral fraud*-. To estimate the substantive effects of my main independent variable, I use the Clarify post-estimation software developed by King, Tomz, and Wittenberg (2000).[146] A key advantage of the Clarify software is that it simulates quantities of interest for the logit models used described before. The graph presented in Figure 4.3 allows us to get a more intuitive understanding of the substantive effects of electoral volatility on my dependent variable. Specifically, Figure 4.3 indicates that the likelihood of *electoral fraud* increases from 1% to 33% when the value of *electoral volatility* goes from its minimum to its maximum and all other control variables in the model are set to their means. This effect is substantially and statistically significant at the 95% confidence level.

Unlike the strong and substantive support that I obtain for my hypothesis 2.3, I find mixed empirical support for the variables that have been identified by other scholars as the determinants of electoral fraud. To begin with, the variable *partisanship* has a negative and statistically significant coefficient (1% confidence level) in models 3 and 4. These results suggest that, holding all the other variables constant, incumbent governments with leftist ideological tendencies are more likely to rig elections. In addition, the positive and significant coefficient, at the 10% confidence level, on *turnout* indicates that the mobilization hypothesis (Ziblatt 2009) finds weak statistical support in my data. In other words, my results suggest that social mobilization, proxied by *turnout*, may indeed positively affect the likelihood of electoral fraud. I lack space here to carefully analyze the mechanisms through which

[146] Substantive effects are estimated using the statistical software Clarify Version 2.0. In short, the Clarify software converts the raw output of statistical procedures into meaningful results for researchers using a Monte Carlo simulation. For more, please see King, Tomz and Wittenberg (2000).

partisanship or turnout affect the likelihood of electoral fraud, but the statistically significant results found here suggest that they are worth exploring in the future.

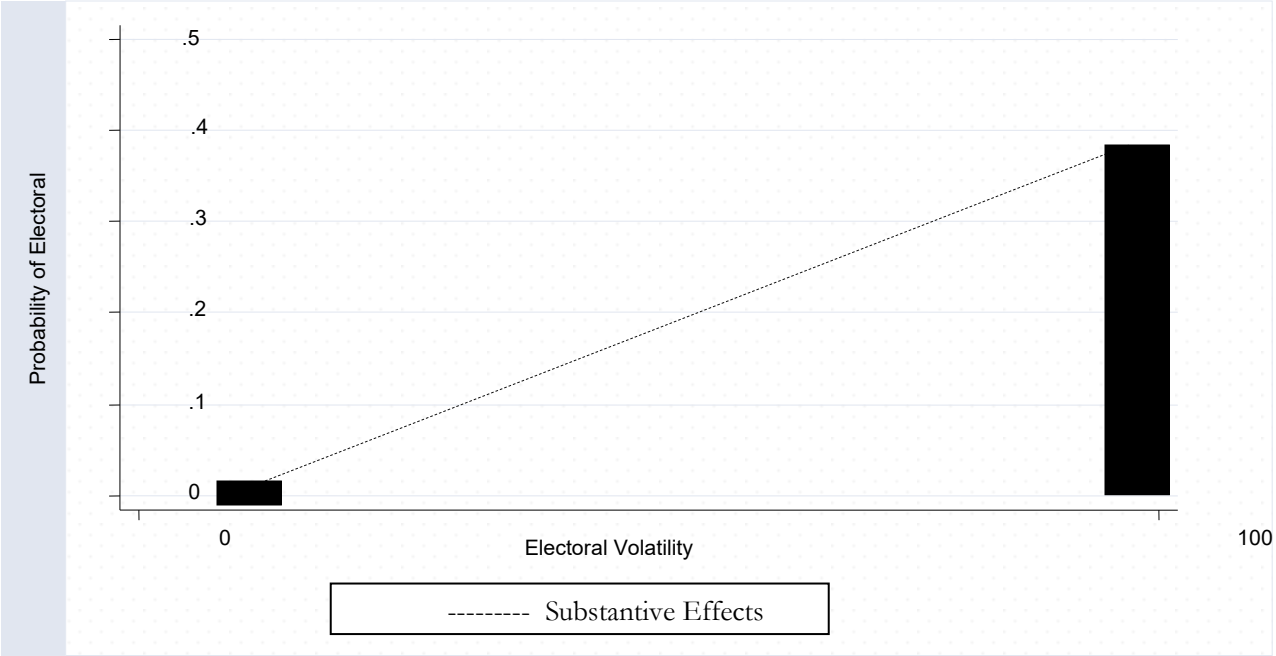

Figure 4.3: Substantive Effects of Electoral Volatility.

In contrast, I do not find statistical support for the alternative hypothesis linking fragile democratic institutions with electoral fraud (Alston and Gallo 2009; Schedler 2006). Although the variables *new democracy* and *EMB* have the expected signs (positive and negative respectively), they lack statistical significance. The coefficient of *polity* is insignificant as well. Likewise, my results do not support previous findings linking electoral fraud with low levels of economic development or urbanization (Auyero 1999; Shaffer 2002; Schedler 2002; Dominguez and McCann 1996; Ziblatt 2009). The coefficients on *log gdp* and *urban* are both insignificant. The studies suggesting that sociological factors largely explain electoral fraud (Trejo and Aguilar-Rivera 2002; Lehoucq and Molina 2002) also lack empirical support in the Latin American region between 1982 and 2006. Both *elf* and *religious* fractionalization are

statistically insignificant. The results of the statistical models presented above suggest that across time and countries it is political and not economic or sociological factors what explains the incidence of electoral fraud in Latin America between 1982 and 2006.

The coefficient on the lagged dependent variable is positive and highly significant at the 1% confidence level in all the models presented in table 4.3, suggesting that occurrence of *electoral fraud* at (t-1) increases its likelihood at time t. All the time transformations are insignificant across my models, indicating that time dependence is not a problem in my data.[147] Although I am confident about the fit of my models, I measure the goodness of fit of model 4, table 4.3, using the Hosmer-Lemeshow (HL) goodness-of-fit test.[148] The results of the HL test lead me to conclude that the model adequately fits the data (77%).[149] The likelihood ratio test confirms the results from the Hosmer-Lemeshow test.

Briefly summarizing, the results in Table 4.3 provide strong support for the hypothesis that high levels of electoral volatility increase the likelihood of electoral fraud across Latin American countries. To ensure that this result is not driven by the econometric technique employed in models 1-4 or the specification of the models, I conducted a battery of robustness tests.

4.4 Robustness Tests

First, I check for the robustness of my results using three additional specifications. To begin with, I asses the robustness of my results using an alternative indicator of the level

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[147] I re-ran all my models without the time dependence variables –*time* $time^2$ *and* $time^3$-. The sign and statistical significance of *electoral volatility* remain unchanged.

[148] For more, see Hosmer and Lemeshow (1980).

[149] This is because the results are not statistically significant.

of institutionalization of the party system. Mainwaring and Torcal (2005: 19) suggest that the level of personalism in electoral campaigns across developing countries is an important indicator of the weak institutionalization of the party system. Following these scholars, I include the variable *outsider* in model 5, Table 4.4. This variable measures the appearance of individual personalities that have a sizable impact in electoral campaigns regardless of their political affiliation. In other words, *outsider* captures the importance of personalism in presidential electoral campaigns across Latin America between 1982 and 2006. I use a dummy variable to operationalize outsider. It takes the value of "1" when the election is won by an outsider candidate and "0" otherwise.[150]

I also add the following several control variables that have been identified in existing studies as determinants of electoral fraud: balance of payment crisis (*bop crisis*), economic crisis (*economic crisis*), electoral competitiveness (*margin*), and income inequality (*gini*) and present the results in Models 6-8, Table 4.4.

Klesner (1993) and Pacek and Radcliff (1995), for example, suggest that rigging election in developing countries is more likely to happen when economic crises occur and domestic economic conditions deteriorate considerably. To assess this possibility, I include the variables: economic crisis (*eco crisis*) and balance of payment crisis (*bop crisis).* Specifically, an economic crisis exists if one of two conditions holds: either the country's inflation rate was 40% or more and it increased by 25% or more from the year before, or per capita GDP fell by 15% or more from the previous year. I label this variable as *economic crisis* which is

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[150] For example, in Brazil, Fernando Collor de Mello won the 1989 presidential election representing the National Reconstruction Party, a centrist political party formed in early 1989. So the variable outsider takes the value of "1" in Brazil from 1989 to 1992. Another example is the 1990 presidential election in Peru, when Alberto Fujimori won the presidency also as an outsider candidate. Additional examples include Venezuela in 1993 and 1998, Colombia and Ecuador in 2002 and Bolivia in 2005.

coded as 1 if either of the two conditions mentioned above is satisfied, and is coded as 0 otherwise. The dummy variable *bop crisis* (where *bop* denotes balance of payments) is coded as 1 if a country's level of international reserves falls to less than the equivalent of three months' worth of imports, and is coded as 0 otherwise. Both variables are taken from Milner and Kubota (2005) and are updated in my data until 2006.

Scholars also suggest that post-election conflicts increase where political competition is more intense (Eisenstadt 1998, 1999; Dominguez and McCann 1996; Lehoucq and Molina 1999; Simpser 2004; Ziblatt 2009). To account for the possibility that the level of competitiveness in elections affects the likelihood of electoral fraud,[151] I introduce the variable *margin* which is operationalized as the difference in votes between the first and second-place winners (Cox 1988). I hypothesize that the narrower the margin of victory, the more likely incumbents are to manipulate elections.[152]

Studying the case of Germany in the nineteenth century, Ziblatt (2009) finds that the level of landholding inequality significantly increases the likelihood of electoral fraud. To evaluate Ziblatt's finding in my data, I introduce the *Gini* coefficient in my regressions. The Gini index ranges from 0 to 1. A low Gini coefficient indicates a more equal distribution, with 0 corresponding to complete equality, while higher Gini coefficients indicate more

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[151] For instance, Dominguez and McCann (1996) argue that accusations of fraud in Mexican elections were more common in competitive districts. Using a sample of more than 300 municipal elections in Mexico between 1998 and 1999, Eisenstadt (1999) finds that competitive elections are positively correlated with accusations of electoral fraud.

[152] I use the following sources to estimate the variable *margin*: (i) The Political Database of the Americas,[152] (ii) International Democracy and Electoral Assistance,[152] (iii) Latin American Network Information Center,[152] (iv) Nohlen (1993), (v) Instituto Federal Electoral (Mexico), (vi) The *Europa World Year Book*, and the (vii) *International Almanac of Electoral History*

unequal distribution, with 1 corresponding to complete inequality.[153] Data for this variable is drawn from multiple sources including Deininger and Squire (1996), the World Bank's WDI (2007), and the United Nations' *World Income Inequality Database*.[154]

TABLE 4.4

PANEL-LOGIT ESTIMATIONS: ELECTORAL VOLATILITY AND ELECTORAL FRAUD IN LATIN AMERICA 1982-2006

[153] The inclusion of this variable reduces the sample by more than 40%, thus I computed a separate model to test the effects of income inequality –*Gini*-.

[154] Since data on income inequality is had to get for some Latin American countries, I tried to maximize the number of observations using several databases. To ensure that the data points are as comparable as possible, I used observations that satisfied the following criteria: (i) they use national representative surveys, (ii) they cover the entire population rather than subsets –rural or urban-, and (iii) they include all types of income.

	Model 5: Logit	Model 6: Logit	Model 7: Logit	Model 8: Logit w/robust s.e.	Model 9: Relogit
Lag Fraud	4.25*** (.713)	8.44*** (2.86)	4.52*** (1.00)	5.05*** (1.28)	3.42*** (1.21)
Electoral Volatility		.072*** (.034)	.084*** (.028)	.092*** (.027)	.063*** (.027)
Outsider	2.23** (1.30)				
Polity	-.172* (.101)	.180 (.282)	.002 (.183)	.052 (.231)	.062 (.219)
New Democracy	2.09*** (1.08)	4.90* (3.06)	2.44 (1.75)	.033 (1.80)	.279 (1.71)
Partisanship	-.968 (.680)	-5.20*** (1.78)	-3.28** (1.89)	-2.58** (1.48)	-1.55 (1.34)
Turnout	-.032 (.028)	.131* (.079)	.018 (.038)	10.7* (5.97)	5.37 (5.67)
EMB		-.167 (.345)	-.143 (.256)	-.373* (.231)	-.157 (.219)
Margin		-.216* (.134)			
Log GDP	-.926 (.885)	-7.05*** (3.37)	.581 (1.77)	-1.79 (2.36)	-1.31 (2.24)
Urban	-1.91*** (.725)	.063 (.095)	-.061 (.058)	-.008 (.048)	-.003 (.045)
BOP crisis		2.20 (2.12)			
ECO crisis		-32.8 (44.4)			
Inequality			-.224* (.129)		
ELF	4.24 (2.09)	9.12 (5.50)	.541 (4.12)	-2.00 (3.00)	-1.82 (2.24)
Religious Fract.	-1.63 (1.87)	2.01 (4.34)	-.628 (3.20)	3.47 (2.35)	1.89 (2.23)
Time	-.277 (.821)	-2.27 (2.70)	-1.36 (1.56)	-.338 (1.01)	-.373 (.961)
$Time^2$	.049 (.077)	.346 (.258)	.185 (.143)	.056 (.106)	.042 (.100)
$Time^3$	-.001 (.002)	-.012 (.006)	-.006 (.003)	-.002 (.003)	-.001 (.002)
Constant	24.01*** (11.09)	42.8 (42.2)	3.38 (14.8)	4.86 (16.1)	6.33 (15.2)
Wald (x2)	145.23	77.90	93.01	83.46	
N	333	290	182	308	298

Model 5, Table 5.5, shows the results of the model that is estimated using the variable outsider as an indicator of the degree of institutionalization of the party system. The coefficient on outsider is positive and statistically significant at the 5% confidence level. In addition, I find that the coefficients on *new democracy* and *polity* have the expected effects on electoral fraud and are statistically significant at the 1% and 10% respectively. Moreover, urban has the expected negative sign and is statistically significant at the 1% confidence level, suggesting that the likelihood of electoral fraud decreases as the percentage of population living in urban areas increases. In sum, the results in Model 5 indicate that the results presented in Table 5.3 remain robust even after using a different indicator of the degree of institutionalization of the party system.

Models 6 and 7, table 5.5, present the results of the specification robustness tests. Specifically, in model 5, I add the variables margin, bop crisis and economic crisis to the full specification presented in model 4 (table 5.3). The results suggest that electoral volatility exerts a positive and statistically significant effect (at the 1% confidence level) on the likelihood of electoral fraud. In substantive terms, the results presented in model 5 suggest that the probability of electoral fraud increases from .06 to significant .39 when the value of electoral volatility goes from its minimum to its maximum and all other variables are set to their means. I then add the variable income inequality to the specifications mentioned above. The sign and statistical significance of the main independent variable, electoral volatility, remain positive and highly significant. Therefore, the results presented in models 1-4, table 3.3, remain robust even when I control for additional variables.

<<Insert Figure 5.4 about here>>

The results in model 6 also provide strong statistical support for hypothesis 2.3. As showed in table 5.5, the coefficient on electoral volatility is positive and statistically

significant at the 1% confidence level. The sign on *gini* (i.e. income inequality) is negative and statistically significant at the 10% level. This result is counterintuitive since I hypothesized that higher levels of income inequality increase the likelihood of electoral fraud.[155]

In addition to specification robustness tests, I check whether my results remain robust when I use different econometric methods to estimate my models. First, the models 1-6 in tables 5.4 and 5.5 are estimated assuming that the errors are normally distributed. Yet, if this assumption is violated, the estimated parameters are likely to be biased and inefficient. The kernel density of the residuals illustrated in graph 3.2 suggests that assuming that the residuals are normally distributed with mean zero may be problematic. To discount the possibility, I thus re-estimate the models using a panel logit estimator with robust standard errors and controls for time dependence and autocorrelation.

Second, the dependent variable –*electoral fraud*- has dozens to hundreds of fewer ones (events) than zeros (non-events). The rare occurrence of the dependent variable, in turn, may lead popular estimation procedures (i.e. logit) to underestimate the probability of rare events. Hence, following King and Zeng (2000), I implement the rare events logit estimator (RELOGIT) to assess the possibility. The RELOGIT procedure estimates the same models estimate the same model as the standard logistic regression, but the estimates are corrected for the bias that occurs when the sample is small or the observed events are rare, which is the case in this study.

Finally, I assess the robustness of my results when the models are estimated via panel probit models with random effects. In contrast with the logit model, the probit model assumes that the error term is normally distributed with a mean of zero and a unit

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[155] Alternative hypothesis based on Ziblatt (2009).

variance.[156] While the choice of which model to use generally is personal, it should be noted that the ratio of the parameter of a logit model to the parameter of a probit model (using the same data set) usually varies between 1.6 and 2.0. For the sake of space, I do not present a detailed description of the results of the probit models.[157] Yet they confirm the robustness of the results presented in Table 5.4 which were obtained using logit estomators.

I present the results of the models estimated using a logit estimator and robust standard errors in model 7, table 5.5. Model 7 has the specification used in model 4 which includes, in addition to the main independent variable, electoral volatility, the lagged dependent variable and controls for time dependence, the variables polity, new democracy, partisanship, turnout, emb, log GDP, urban, elf, and rel_frac. The sign on the coefficient of electoral volatility remains positive and highly significant (at the 1% level), indicating the robustness of my previous findings. Substantively, this result suggests that the likelihood of electoral fraud increases from a substantial 17% to 44%.[158] As I previously found, the coefficient on partisanship is negative and statistically significant at the 5% confidence level. Turnout is positive and statistically significant at the 10% level and EMB has the expected negative sign –i.e. more independence of electoral management bodies leads to less likelihood of electoral fraud- and is statistically significant at the 10% confidence level. The results of model 7 do not support the hypotheses advanced by other scholars linking economic and sociological factors with electoral fraud in the developing world (Auyero

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[156] The assumption that the variance is equal to 1 is due to technical considerations (Cramer 2003).

[157] These results are available upon request.

[158] This is when electoral volatility goes from its minimum to its maximum and all other variables are set at their means.

1999; Trejo and Aguilar-Rivera 2002; Lehoucq and Molina 2002; Shaffer 2002; Schedler 2002).

Model 8, table 4.4, report the results of the RELOGIT estimator. The coefficient on electoral volatility remains positive and statistically significant at the 1% confidence level. Although estimating the substantive effects of the main independent variable is not possible using the RELOGIT estimator, the positive and significant coefficient on electoral volatility suggests that the results reported in models 1-4, Table 4.4, are robust even when the model is estimated via the rare events logit estimator. The lagged dependent variable is also positive and significant (1% confidence level), but the additional control variables lack statistical significance.

4.5 Conclusion

The frequency and pervasiveness of electoral fraud in developing countries and its potential negative effects for democratic stability and survival has motivated social scientists to increasingly study the causes that fuel this phenomenon (Auyero 1999; Eisendstadt 1999; Alston and Gallo 2009; Ziblatt 2009; Lehoucq 2003; Lehoucq and Molina 2002). These studies have advanced our understanding of electoral fraud in developing countries but they suffer from two important drawbacks. First, most of the existing research suggests that economic and sociological factors are the main determinants of electoral fraud, ignoring that democratic political institutions could have an important effect on this act of political corruption.[159] In addition, the lack of reliable data on electoral fraud has prevented scholars from assessing the validity and generalizability of their claims in a cross-sectional setting. In

[159] Lehoucq and Molina 2002 and Lehoucq 2003 are some exceptions.

this chapter, I avoid these problems estimating the effects of a political variable (i.e. the level of institutionalization of the party system) on electoral fraud using a comprehensive time-series-cross-sectional dataset of 18 Latin American countries between 1980 and 2006.

I find that high levels of electoral volatility (used as a proxy for the level of institutionalization of the party system) increase the likelihood of electoral fraud. In substantive terms, my results indicate that the likelihood of *electoral fraud* increases from 1% to 33% when the value of *electoral volatility* goes from its minimum to its maximum and all other control variables in the model are set to their means.[160] Therefore, my results strongly support the claim posited in hypothesis 2.3 –that low levels of party system institutionalization increase the likelihood of electoral fraud.- Interestingly, my results also suggest that partisanship and turnout also influence the probability of electoral fraud across Latin America. In contrast to previous studies, I don not find any empirical evidence supporting the alternative hypotheses that link electoral fraud with economic and sociological factors.

Two important substantive conclusions emerge from this chapter. First, much of the existing literature on the determinants of electoral fraud focuses its attention in explaining the existence of this phenomenon using economic and sociological factors (Auyero 1999; Trejo and Aguilar-Rivera 2002; Alston and Gallo 2009; Eisendstadt 1999; Ziblatt 2009). However, the empirical analysis presented here suggests that political factors better explain the likelihood of electoral fraud in Latin America between 1982 and 2006. More specifically, the results presented here suggest that a key characteristic of the party systems in Latin America, namely the degree of party system institutionalization substantively affects the

[160] Substantive effects were estimated using the results from Model 4, Table 3.x.

probability of this act of political corruption. Therefore, my results suggest that scholars should devote more time and effort to carefully analyze, theoretically and empirically, whether other political institutions affect the likelihood of electoral fraud and to what extent.

Second, the results presented in this chapter confirm the intuition that weak party system institutionalization has important consequences for democratic politics (Mainwaring and Scully 1995; Moser 1999, 2001; Stoner Weiss 2001; and Mainwaring and Torcal 2005). Yet they only provide empirical support for one (i.e. electoral fraud) of arguably many different causal mechanisms through which party system institutionalization may affect democratic politics. I lack space here to analyze other potential mechanisms but such research agenda is worth pursuing.

GNU GENERALISMS

A.1 Definitions

Several definitions are presented in Table A.1 to show both how to do rotated, line-spanning tables, as well as to define some commonly used Gnu terms.

TABLE A.1

COMMONLY USED GNU TERMS

Term	Definition
Gnu	Small furry animal that is related to the squirrel (although they won't admit it).
LoG	Abbreviation for the "Leader of Gnus". See Chapter 2.
Twizzlers	Red, twisty candy that is among the most favorite of Gnu foods. Gnus frequently appear overly cute and friendly to humans bearing twizzler packages. This is known as "trolling for twizzlers" among the Gnus.

Finally, Table A.2 shows the top ten Gnus from Table 2.1 ranked in order by their aggregate score (along with some of the raters' comments). This follows a long-standing Gnu tradition of self-improvement through public announcement of score (which some associate with military origins [1]). Indeed, this very table has been observed in the Gnu lodge where it was posted for peer review.

TABLE A.2

ALPHABETICAL LIST OF TOP TEN GNUS FROM TABLE 2.1,

WITH REVIEWER COMMENT

Candidate	Aggregate Score	Reviewer Comments
George	7.6	George is an excellent candidate for the LoG. Slightly low C, but hopefully, this 7.6 will be high enough!
Glen	7.2	A little weak on AM and Fr, but good scores overall. One or two more years of experience should be enough.
Goldie	7	Dismal score in Fr; suspect it had something to do with strenuous weight loss program this past year.
Gillian	6.9	Excellent C, but a little shabby on the Fu. Suggest more roughage.
Gibby	6.9	Reasonable scores, but need to work on Fr. Gibby is definitely not a morning Gnu.
Genaveve	6.5	Very low Fr; perhaps more coffee? Suggest practicing "cute faces" in the mirror several hours per day.
Giovanni	6.2	Very low Fr; suspect hanging out with Genaveve too much.
Gina	6.2	Mediocre Fu, somewhat low AM. Perhaps a future in marketing or advertising?
Garrick	6.2	Fairly low AM. Fu could be better as well; buy a comb. And a mirror. Immediately.
Gardenia	6	Dismal AM; very low Fu. Seems to care more about meeting agendas than personal appearance.

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